

PROJECT MANAGEMENT INFORMATION SYSTEM FOR

MODULE PRO FURNITURE AND FIXTURE

A Thesis

**Presented to the Faculty of
Information and Communications Technology Program
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**In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Computer Science**

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ABSTRACT

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This project is centered around designing and developing a Project Management Information System (PMIS) for Module Pro Furniture and Fixture, a small business that specializes in building custom furniture and fixtures. The PMIS will facilitate the company's goal of updating its manual operating procedures, which include tracking project expenses, payments, and scopes of work, to a more accurate, organized, and productive digital form. The PMIS was created utilizing C# (WinForms) and SQL Server Express while also utilizing the Kanban Agile Model to ensure continual progress and adaptation to changes over the course of the development cycle. The PMIS will include key features such as project creation, project payment tracking, timelines of completion, and notification automation to facilitate payment collection. The system will also provide user access retention for data integrity and to ensure certain operations can only be performed by assigned users.

The results showed that the system worked well and matched the client's operational requirements. The PMIS has not only taken the place of manual recording methods but also provided automated modules that reduced human error and offered improved financial management power and coordination among staff and administrators. The PMIS shows that even smaller enterprises can make a meaningful difference with organized systems that provide operational efficiency, data accuracy, and better decision-making in project management.

TABLE OF CONTENTS

	Page
Title Page	<i>i</i>
Endorsement Form for Proposal Defense	<i>ii</i>
Approval Sheet	<i>iii</i>
Acknowledgments	<i>iv</i>
Abstract	<i>v</i>
Table of Contents	<i>vi</i>
List of Tables	<i>vii</i>
List of Figures	<i>viii</i>
List of Appendices	<i>ix</i>
Introduction	1
Background of the problem	1
Overview of the current state of technology	2
Objectives of the study	4
Scope and limitations of the study	5
Review of related literature/studies/systems	22
Methodology	28
Results and Discussions	32
Conclusion	34
References	35
Appendices	37

LIST OF TABLES

Table		Page
1	Gantt Chart	39

LIST OF FIGURES

Figure		Page
1	Sample Project Details	100
2	Material Details of the Sample Project	101
2.1	2nd Page of Material Details of the Sample Project	102
3	Kanban Agile Methodology Model	104

LIST OF APPENDICES

Appendix		Page
A	Gantt Chart	38
B	Actual Thesis Expenses	40
C	User's Manual	43
D	Admin's Manual	45
E	Curriculum Vitae of Researchers	47
F	Calendar of Activities	52
G	Transcript of Interview	94
H	Documented Images	99
I	Kanban Agile Methodology	103

INTRODUCTION

In the competitive landscape of custom furniture businesses, effective project management plays a crucial role in ensuring timely delivery, cost control, and customer satisfaction. Managing multiple projects simultaneously while keeping track of budgets, sales, purchase history, and overall project expenses can become overwhelming, especially for small and medium enterprises (SMEs). These challenges often lead to delays, financial mismanagement, and inefficiencies in day-to-day operations, which can negatively impact both profitability and client relations.

This paper proposes the development of a tailored Project Management Information System (PMIS) for Module Pro Furniture and Fixture, a custom furniture business. The goal of this system is to streamline the management of the company's projects by providing an organized framework where all critical information, including project budgets, sales, purchase history, and billing, is consolidated into one platform. By integrating these functions, the system will not only improve efficiency but also enable better financial oversight, ensuring that each project is completed within budget and according to the client's specifications.

Through this project, the proponents aim to address the growing need for an orderly and transparent method of handling multiple custom furniture projects, offering a solution that simplifies complex processes and supports decision-making. The system will provide real-time data on financials and project progress, enabling managers to make informed decisions and ultimately drive the business toward greater efficiency and profitability.

Background of the problem

Module Pro Furniture and Fixture, led by Mr. Rimbart Romero, is a startup business that offers custom furniture, cabinets, and modular design for home renovation purposes and other client-specific requirements. For almost two years, it has been up and running and appears to be gaining traction among its clients (*see Appendix G, pg. 95, q. 1*). However, Mr. Romero presents several challenges that hinder the company's growth and efficiency.

Currently, the company relies on manual methods for managing financial transactions such as recording project expenses, project budgets, and payments. Mr. Romero currently uses notebooks and, occasionally, MS Excel to log key information such as project expenses, and material purchases (*see Appendix G, pg. 95, q. 3*). This approach is inefficient and prone to errors as these issues cause difficulties when it comes to tracking project-specific budgets, assessing remaining funds, and ensuring proper allocations of resources, which all lead to overspending or delayed project completion.

Additionally, according to Mr. Romero, he occasionally forgets to record the data due to multiple projects being handled simultaneously. Managing multiple projects simultaneously often leads to lapses in recording and tracking vital data, such as material costs, client payments, and bill payments (*see Appendix G, pg. 95, q. 5*). These lapses disrupt financial transparency and create discrepancies that will make it challenging to monitor expenses and the overall assessment of the company's financial status.

In addition to Mr. Romero's occasional forgetfulness of recording due to simultaneously handling multiple projects, according to Mr. Romero, handling multiple projects also causes him to be "off-track", he needs to be reminded that he must pay the necessary expenses (*see Appendix G, pg. 95, q. 5*). The absence of an automated system to remind Mr. Romero about due dates for payments and bills further exacerbates the problem. Missed deadlines can lead to delays in supplier payments and missed opportunities to follow up with the clients, which negatively impact the company's operations and positive relationships with the client.

Overview of the current state of the technology

In today's business landscape, especially for small and medium enterprises (SMEs) like custom furniture businesses, managing multiple projects while keeping track of sales, purchase history, and project expenses has become increasingly complex. Many businesses still rely on manual methods or generic project management tools, which are not tailored to the specific needs of industries like furniture and fixtures, leading to inefficiencies and financial oversights.

Currently, several solutions exist to address parts of these challenges, but none of them fully integrate the unique aspects of managing a custom furniture business. Most businesses use separate systems or tools for project management, financial tracking, and sales reporting, which can lead to information silos, where data is not easily accessible across departments. This fragmentation results in delays, increased administrative burden, and the risk of financial mismanagement.

Commonly Used Solutions

Several existing solutions are being implemented by businesses to handle these issues, but each has its limitations:

Manual Record-Keeping Systems

Many custom furniture businesses, just like Module Pro Furniture and Fixture, still use spreadsheets or paper records to track sales, purchases, and bills (*see Appendix G, pg. 95, q. 3*). While this approach is simple and inexpensive, it is highly prone to errors, is time-consuming, and does not provide real-time updates on project finances.

General Project Management Software

Although Mr. Romero did not use the following tools, like Microsoft Project or Trello, these are still commonly used for managing tasks and timelines, but lack integrated financial tracking features, such as sales history or bill management. These tools are effective for project planning but require supplementary tools for financial control.

Basic Accounting Software

Although Module Pro did not use the following accounting programs like QuickBooks or Xero, these programs are still popular for handling financial transactions, sales, and purchases for other businesses. However, they are primarily designed for general accounting purposes and are not optimized for managing multiple custom projects, which often require more detailed tracking of materials, labor costs, and ongoing expenses per project.

Limitations of Current Solutions

Despite the availability of these tools, the current state of technology presents several limitations:

- Lack of integration between project management and financial tracking, leading to data being spread across multiple systems.
- Manual systems are prone to errors and inefficiencies, especially when handling large volumes of data for multiple projects.
- Existing tools do not cater to the specific needs of custom furniture businesses, where tracking individual project expenses, materials, and labor costs is essential.
- Financial data, such as sales history, purchase records, and bills, is often not connected to project timelines, making it difficult to get a full overview of each project's financial health in real time.

Objectives of the study

This study aims to develop a Project Management Information System (PMIS) tailored to the needs of Module Pro Furniture and Fixture, a custom furniture business. Specifically:

- 1. To design and develop a module that facilitates efficient tracking of project expenses.**

The developers will create a module that will allow the client to record, monitor, and manage project-related financial transactions in an organized and automated manner. This includes recording project-specific expenses and material costs. The module will feature the categorization of data to ensure that all expenses are logged under the appropriate project and a project breakdown to avoid overspending and better allocation of resources. This module reduces the risk of errors, improving financial clarity.

2. To design and develop a module for streamlined recording and tracking of customer payments and material purchases.

The developers will create a module that enables the client to efficiently record and track all customer payments and material purchases of the client for each project. This includes client payment and material purchases. The module will check payment records, considering the material purchases, providing an overview to the client, and avoiding any discrepancies between what has been paid and what is due. This ensures accurate documentation and reduces lapses or human errors due to manual entry.

3. To implement an automated notification system to prevent missed payments and ensure timely financial management.

The developers will create a module that allows notification to the user if there are payments to be paid, based on the given due date of the said payment. The notification will not be limited to the platform itself; rather, the client will also be notified on the user's phone via email. This module will help the client stay on top of due dates for bill payments, which minimizes disruptions caused by oversight in managing multiple projects simultaneously.

Scope and Limitations of the Study

Scope of the Study

The system aims to address the business's need for a more organized and efficient approach to handling multiple custom furniture projects. The target user of this system is mainly Mr. Romero the business CEO, as a solution to the problem he is currently facing, who will benefit from having all relevant project/s information organized and managed in one place. This system will allow them to monitor projects, project budgets, track sales, and manage and overview purchase records, and review bills and expenses in real-time. By integrating these features, the PMIS is designed to improve operational efficiency and financial oversight.

Front-End Modules

- **Staff User-Access Level**

- Creation and requests of projects will be handled by the staff of Module Pro Furniture and Fixture. The staff user can access the following modules:

- **User Dashboard**

- The User Dashboard serves as Module Pro's central hub, providing a clear overview of pending, active, and finished projects. It includes a notification tab for project status updates and showcases the top five most selected scopes within each relevant section. This is to offer a centralized, intuitive space for the user/s to quickly access important information, track project progress, and engage with the system.
- **Notification:** This notifies the staff about the results of the requests they have made from a project - whether a request has been approved or rejected along with its reason. This also notifies the staff about the statuses of a project, i.e. is about to start or when the project is still active yet the expected end date is near. A "Mark all as Read" button is available to hide notifications that have already been read.
- **Top Selected Scope of Work:** This displays the top 5 most selected scope of work regardless of the project area. It shows the scope of work's name and the amount of times that scope of work is selected. This provides insights for the staff which scope of works are mostly chosen by the customers.

- **Pending Projects Nearing Validity:** This displays the top 5 pending projects that are nearing the quotation validity of an unpaid project. This provides awareness to the staff so that they can contact the customer about the status of their project.
- **Active Projects Nearing Start:** This displays the top 5 active projects that are going to start either by this month or so on. This provides knowledge to staff so they are well informed that a project is going to start.
- **Recently Finished Projects:** This displays the top 5 recently finished projects. This is only to inform the updated status of the finished project so that the staff can then inform the customer about the finished status of their project.

- **Projects Dropdown**

- This dropdown menu shows all the projects categorized by their project status: Pending, Active, and Finished. Alternatively, the staff can also see All Projects, excluding cancelled and archived projects. Each module instantiates their own independent table, each of which is clickable which then shows comprehensive details about the selected project.
- This also incorporates a filter option, which allows users to efficiently organize and refine displayed data according to specific criteria, thereby enhancing accessibility and usability. By having well-managed and categorized projects, this will help the staff navigate the projects easily. The following are the modules of the dropdown menu that are categorized by their project status:

- **Pending Projects:** It displays all the pending/for approval projects via table that shows a summary or small information about the project. Each row is clickable, allowing the user to view the full details. The module features a search function - search by project name and client name - and a sorting feature - quotation validity, unpaid, partial, and fully paid.
- **Active Projects:** It displays all the active projects via project table that shows a summary or small information about the project. Each row is clickable, allowing the user to view the full details. The module features a search function - search by project name and client name - and a sorting feature - start date, end date, unpaid, partial, and fully paid.
- **Finished Projects:** It displays all the finished projects via project table that shows a summary or small information about the project. Each row is clickable, allowing the user to view the full details. The module features a search function - search by project name and client name - and a sorting feature - recently ended and old finished projects.
- **All Projects:** This shows all of the projects excluding the cancelled and archived. The module only features a search function - search by project name and client name.

- **Portfolio**

- This provides a collection of images of all finished projects including the archived projects, which is used for examples if the customers wish to view previous works.

- Each card is clickable, which shows all the details about the projects. It also includes a search bar that allows filtering by client name, project name, or project area. This helps the business present completed works while making it easier for the customers to explore past projects.
 - This is an excellent tool in enhancing sales and establishing customer trust in Module Pro Furniture and Fixture.
- **Project Creation**
 - This three-page module enables the user to create a project by clicking the "New Project" button. The creation process is split into three pages: Area Selection, Scope of Work Selection, and Basic Details. Based on Mr. Romero (*see Appendix G, pg. 96, q. 6-9*), the following information must be entered to create the project:
 - **Area Selection**
 - **Project Area:** selection of area. User can only select one (1) area for the project.
 - **Scope of Work Selection**
 - **Scope of Work:** Selection of scope. Users can select one (1) or more for the project. Module also allows the user to see the price of that scope of work and can modify how many units the customer wants.

- **Basic Details**

- When creating a new project for a new client, the email and contact number must be unique and not match any existing client information.
- For existing clients, their basic details can be filled automatically by pressing the Tab key, excluding the project description. These details can be edited and any changes require confirmation.
- **Project Name:** An identifier for the project. This is auto-generated upon submission.
- **Client Name:** The full name of the client.
- **“Reset”** button. This helps the user if they accidentally selected an existing customer. Clicking the button clears all client input fields, including the selected existing customer.
- **Client Email:** An optional input of the customer’s email.
- **Client Contact Number:** The primary phone number of the client for communication purposes.
- **Client Address:** The place of the client or the project’s location.
- **Project Description:** A brief overview of the project, highlighting its purpose, the

client's idea or concept, and the overall scope of the project.

- This module helps the business stay updated and on track with its projects, the Project Creation module ensures that all critical details are captured, enabling efficient monitoring and management.
- After filling up the necessary information of the project, all the information will be displayed once the user click the “Proceed to breakdown” button.
- **Project Breakdowns**
 - There are two modules in this category and is shown depending on the project's status:
 - **Project Proposal Form**
 - This is essentially the last page of the project creation module. This module shows all the information that was entered in the project creation module. Additionally, it will also show a calculated cost breakdown of the project - the project cost, downpayment, and progress billing cost. It will also show the quotation validity date (valid until) - a date of which if a project is left pending and unpaid, it will automatically be cancelled.

- **Project Details**

- This module opens when the user clicks a project card. This shows all of the information about the project, just like the Proposal Form, however, additional information such as the assigned start date, approximate end date, project work area, attention to, project and payment status, project and payment history timeline, completion proof, and cancellation reason, are shown to the staff. To update the payment and project status staff can also make requests with a click of a button. Additionally, to provide a better connection with the customer, the staff can print the displayed information so that the customer is aware of their project's status and information.
- *Note:* Completion proof and cancellation reason are not shown simultaneously. They are dependent on the project status.

- **Project Request Module**

- This sub-module, which will show upon clicking a button in the project details, features a dropdown list of four (4) types of requests that the staff can make that will help update the status of a project as well as its payments.

- **Payment Update Request**

- This selection provides the financial information of the selected project along with an input field for the staff to enter the amount paid by the customer. For unpaid projects - the staff can only input the exact down payment amount or the exact project's cost. The staff or customer should not pay higher than the conditions nor lower as according to the company's terms and conditions they are calculated at a fixed rate. For Partially Paid projects - the staff must input equal or lower than the progress billing or the outstanding balance. Unpaid and partially paid projects are allowed to make payment requests.

- **Completion Request**

- The staff must insert an image showing the proof that the project is indeed completed. Only fully-paid active projects are allowed to make this request.

- **Delivery Request**

- The administrator is permitted to approve delivery requests only after confirming that both the corresponding payments and project completion requirements have been satisfied.

- **Cancellation Request**

- The staff must only input a reason for cancellation before submitting. This request is only allowed for unpaid or partially paid pending or active projects.
- The request button will be unavailable for fully or partially paid pending projects - the project is in “waiting” state as it requires the admin to approve the project - and fully paid finished projects

Back-End Modules

- **Authorization and Security Access**

- This module controls user access levels, ensuring that sensitive information is only available to authorized personnel. It includes login authentication and security protocols. Additionally, it includes an account creation module where the admin can create and label the user to their respective user access. Login modules require the user’s username and password. In the User Management Module the admin should include additional information which is the input of their email, as this will help for the notification payment reminder module. Staff or other users, on the other hand, only require their preferred username and password for the account creation.
 - To safeguard data integrity, prevent unauthorized access, and protect both the company and client information through security measures.

- **Administrator User-Access Level**

- **Analytical Dashboard**

- A data-driven interface that presents key business metrics, trends, and performance indicators, specifically for internal team use. It

may include visualizations like graphs and charts.

- To assist management in making informed decisions by providing a comprehensive view of business performance and project data.
 - **All Time:** This panel shows the status of all projects, including its total number of projects.
 - **Notifications:** This notifies the admin about all due payments and requests from staff, including requests for payment update, completions, delivery, and cancellations. This ensures the admins are informed ahead of time about schedules and deadlines, reducing the risks of delays especially in payments.
 - **Actual Gross Profit:** This shows a line graph based on paid projects, regardless of project status. All partially and fully paid projects are shown, except archived projects. This data can be filtered by custom ranges, all time, last 7 days, last 30 days or this month.
 - **Donut Graph (Total):** This chart visualizes the project status distribution.
- **Payment Reminder**
 - This back-end module helps the scheduling and sending of payment reminders to the user via email. The reminders are generated based on due dates entered in the system and follow a custom schedule - the user can enter a number and select the desired time (e.g., day/s, week/s, and month/s) before the due date. After generating a reminder from the payment reminder module, an event calendar will be generated and sent to the user's registered email. The user can then open the event from the email on their phone, which then can be imported to the installed calendar app.

- To support timely payments by automating reminders, minimizing manual follow-up, and enhancing operational efficiency.

- **Projects Dropdown**

- A dashboard showing all the company projects, helping Module Pro see active and completed projects in an organized way, which helps the client look at each project overview with ease. This is to offer a centralized, intuitive space for the user to quickly access important information, track project progress, and engage with the system.

- **Pending** - This contains all the pending projects with partial and unpaid payment status. The admin can decide to initiate a project by clicking "accept project." Each row displays important details and is clickable to view the full details. A project can only be accepted or approved if the project is partially or fully paid.
- **Active** - This contains all active projects. Each row displays important details and is clickable to view full information.
- **Finished** - This contains all the finished projects. Each row displays important details and is clickable to view the full details.
- **All** - This shows all the projects: pending, active, and finished, excluding those that are cancelled and archived.
- **Cancelled** - This contains all the cancelled projects including the reason why the project is cancelled.
- **Archived** - This contains all the archived projects including

cancelled projects. This is to maintain uncluttered workspace for active projects, protect older data by hiding it from regular views, and prevent irreversible loss of information by offering an alternative to permanent deletion, thus ensuring data remains retrievable when needed.

- A Search tab is provided for easier navigation of the project. Additionally, a filter tab with dropdown menu is available for more efficient sorting.
- These modules work the same way as the staff, each respective module has their own respective search and sorting features. This provides an easier navigation to an already filtered module.

- **Requests**

- This module displays all requests originating from the staff. It is divided into four (4) panels:
 - **Payment Requests** - This contains all projects with payment updates.
 - **Completion Requests** - This contains all projects that are ready for completion.
 - **Delivery Requests** - This contains all projects that are ready for delivery. Delivery requests require projects that are fully paid and that meet all requirements for delivery.
 - **Cancellation Requests** - This contains all projects that are for cancellation.

- **Project Breakdowns**

- There are two modules in this category and is shown depending on the project's status:

- **Project Approval Form**

- A module that is only shown for pending projects regardless of the payment status. This is the module where the admin first goes to double check the project before approving and making the project active. This provides all of the information from the Project Proposal Form along with additional information that is for the admin only. Additional selection and information such as, the selection of starting date and assigning the attention to, and more details about the selected scope of work like the materials cost and the calculated gross profit, are shown in this module.

- **Project Details - Admin**

- This module essentially functions the same as the Project Details in the Staff but with more financial information about the project's scope of work, specifically, total materials cost and calculated gross profit.
- **Generate receipt** - Generates receipts for the user or customer's awareness, which only the admin can do.

- **Project Settings**

This contains the contents of the scope of work which are the scope areas, scope items, and the scope of work itself.

- **Scope Area**

- This contains areas that allow the admin to select a specific area for the scope of work.

- **Scope Items**

- This contains a list of items that allows the admin to select specific components/items relevant to the scope of work.

- **Scope of Work**

- This contains all the scope including its price, items, and its specification. Admin can select and update the information needed for the scope of work.

- **Attentions**

- This contains all the available engineers, including their phone number. Admin can add, select, and update the engineers and its information relevant for the scope of work.

- **Settings**

- This module contains the project's financial settings, project parameters, and timeline settings, including the quotation validity. This allows the admin input desired parameters, automating calculations and significantly reducing the time and effort that is typically manually computed.

- This helps the admin monitor the project's financial status,

and the project's validity.

- **Backup and Restore**

- A sub-module within the Settings module that backups all critical data related to the company's projects, client information, and operational details. Additionally, a restore module to retrieve the backed-up data.
 - To prevent data loss by regularly backing up essential data, and restoring backed-up data to ensure continuity and reliability in case of technical issues or cyber threats.

- **Users**

- This module allows the admin to manipulate data, create new users, and monitor the status of each user. This module provides a table that shows all the information of each user.

- New user

- **Username** - a unique identifier for each user.
- **Email** - user's email address should be available for contact purposes and for admin, it is also used when generating reminders.
- **Password** - users must create a valid password to create an account. The system provides a correct guideline for creating a valid password.
- **Select Role** - There are two(2) roles for the system's users: Admin and Staff.
 - After putting all the required information for user creation, a confirmation button is

available, and a cancel button for if the admin wants to discard all entries.

- **Search Bar**

- This is a real time search bar that allows the admin to easily navigate users by just typing the username.

- **Filter**

- A dropdown feature that allows the admin to filter the users according to roles for efficient sorting. There are three(3) selections: All, Admin, and Staff.

- **Logs**

- This module contains overall activities of both Staff and Admin. This provides a comprehensive read-only record of all the activities done by the staff and admin including logins, project creations, and requests.

- **Search Bar**

- A real time search bar that allows the admin to easily navigate specific activity.

- **Filter**

- A dropdown feature that allows the admin to filter logs according to the activities which is an efficient way of sorting logs.

Limitations of the Study

- **Payment Processing**

- Integrating a secure and reliable payment processing system requires significant compliance with financial regulations,

additional security features, and partnerships with payment gateway providers. Given the project's focus and resource limitations, payment processing was deemed outside the immediate scope. However, customer payments, purchases, and other relevant financial information that are required for monitoring the financial status of a project will accept cash-only payments, only for monitoring purposes.

- **Delivery Area**

- The delivery and installation fee percentage of a project is not dependent on the distance of the project location. It is calculated based on the total cost of the items of the selected Scope of Work. Since the business currently delivers only within Quezon City and some selected areas of Metro Manila (*see Appendix G, pg. 97 , q.18*), the system can only provide a fixed percentage rate for delivery and installation. However, it can be modified by the administrator via Setting Module; the update rate will then be applied only to projects created after the change.

Review of related literature/studies/systems

Local Literature

According to Ansoos et al. (2020), the use of accounting information systems has been studied concerning the needs of Philippine SMEs, and their advantages in improving project-based financial monitoring have been underscored. Accounting information systems make it easy for SMEs to record and monitor the activities, budgets, and client expenditures across several project undertakings with less effort. They also simplify the presentation of the income and expense amounts to the managers, enabling them to understand very well the financial position of the different projects they are managing. The research further asserts that accounting information systems improve clients' revenue management by enhancing the speed of preparing a project financial synopsis, thus easing the revenue projections.

As Philippine businesses face the challenges of BIR tax compliance, integrated accounting systems seek to provide a solution by incorporating tax modules into the entity's business processes. The discussion here explains how, in doing so, the incorporation of these tax features into accounting software not only reduces the incidence of tax computation errors but also allows for efficient and, above all, timely filing, which is critical for small and medium-sized businesses that have very little time to waste waiting for tax issues. Furthermore, it emphasizes how such technology reduces work drudgery, allowing more effort to be directed toward project growth and timely delivery. The integration of tax compliance modules results in increased precision of tax returns submitted and a better understanding of financial operations, as noted in the research. Angeles (2021).

The engagement of accounting information systems through project tracking abilities has been beneficial, particularly to Filipino firms that have to deal with many projects at the same time. Ramos (2023) highlights that these systems incorporate managing project task timelines, measuring the actual project performance, sales, and purchase history recording, and thus effectively monitoring all the projects. It also emerged that previously untapped benefits, such as the increased ability to modify and manage project budgets, are possible due to the introduction of project tracking alongside financial figures.

Related Systems

Taxumo facilitates the filing of taxes for all businesses registered in the Philippines. Taxumo enables automatic compliance with the BIR and accounting for projects. The system facilitates the tracking of sales, purchases, and payroll, enabling small and medium-sized enterprises to prepare financial information for each project. The tax automation and project-based tracking feature makes it easier to comply with tax requirements for such businesses by allowing the businesses to concentrate on finishing projects and catering to clients without worrying about being compliant with tax regulations.

JuanTax is a tax compliance software based in the Philippines that automates the filing of the Bureau of Internal Revenue (BIR) forms and offers additional accounting

functionalities, such as tracking on a project basis. JuanTax is designed to meet the needs of small and medium-scale enterprises. Within also includes expenses and revenue recording, sales, and purchase histories, as well as tax returns preparation, making it appropriate for organizations that do several client works and are obliged to account for tax purposes and accurate finances.

EZBooks is an accounting program designed for local companies in the Philippines with a primary emphasis on compliance with BIR rules and managing finances on a project level. This software has modules enabling sales, purchase, payroll, and expenditure monitoring in a project-centric approach. It is most suitable for local small and medium enterprises that require ease in carrying out accounting processes, regulating finances by a given project, and maintaining proper tax records. The reports prepared by the program that conforms to BIR regulations help in the efficient preparation of tax returns, which is the main reason the EZ Books program is loved by small-scale business owners in the Philippines.

Foreign Literature

According to Boonpenge (2020), there is a complex relationship between management and accounting systems and corporate performance research. The research was done in Thailand and talks about management accounting applications. Although traditional management accounting practices are commonly employed, there are still inconsistencies in how these systems impact the outcome of business. Studies often lack comprehensive evaluations of the actual effects on business performance, which this research highlights. While the factors influencing management accounting adoption are well-documented. This suggests that while management accounting tools are foundational, they may not directly translate into improved corporate results due to varying implementation and contextual factors.

According to Aliamutu et al. (2020), research in South Africa reveals that many small enterprises face challenges in implementing systems effectively, like accounting information systems (AIS) for enhancing small business profitability. The study emphasizes that having a successful AIS adoption could potentially address operational

inefficiencies and contribute to the financial health and growth of small businesses in the region, as having a lack of effective implementation results in low operational efficiency, which underscores the need for better integration of AIS within small businesses to improve performance and profitability.

According to Fuxing (2020), recent research on enterprise logistics cost management in China has examined various strategies for effective inventory control within intelligent supply chains. The study breaks down logistics costs and emphasizes the importance of tailored cost control mechanisms suited to today's economic conditions. Fuxing argues that the successful implementation of these strategies is essential to minimize operational disruptions and manage expenses efficiently, highlighting that precise logistics cost management can significantly contribute to overall cost reduction and supply chain stability.

Related Systems

Zoho Books is an accounting software solution that provides a variety of financial management features, such as expense management, invoicing, inventory control, and tax processing. Additionally, Zoho Books allows for project-based accounting, where users can track revenue and costs that relate to specific projects. Its tight compatibility with other Zoho applications makes it more appealing to small and medium-sized businesses all over the world.

Xero is an online accounting software ideal for small to medium enterprises as it offers invoicing, expense management, payroll services, and project tracking capabilities, among others. Its guidelines are simple, and Xero allows strong integrations, making it easy for business owners and accountants to collaborate and access up-to-date financial information. This software is adopted in many countries around the globe as it allows trade in different currencies, an advantage for worldwide businesses.

Tally ERP was developed in India but is now focused on Asia and the Middle East in accounting, stock control, and payroll management. The software helps manage advances and is equipped with features like multi-currency, inter-branch, and tax provisions,

thereby it suitable for use by multi-faceted organizations across borders. Its modular structure enables tracking of costs and incomes for each project, making it suitable for small and large organizations.

SAP is an enterprise resource planning (ERP) software that combines core business processes, including finance, supply chain, procurement, and human resources. It offers real-time data analysis, automation, and centralized data management. The system shows scalability, which means it supports various organizational sizes. SAP ERP's modular design allows companies to adopt only the features they need while ensuring seamless collaboration across departments. Widely recognized for its adaptability, it enhances decision-making and operational efficiency, empowering businesses to meet dynamic market demands globally. Its compatibility with technologies like IoT and AI makes it a leader in innovation within the ERP space.

Synthesis

Project Management Information Systems offer significant benefits, especially to a startup company like Module Pro Furniture and Fixtures. Three local studies point out how the systems of project management make financial tracking simple, better the management of revenue, and monitoring of projects for SMEs. Studies in this region target the reduction of manual errors and the ability to provide accurate financial reports, as it is important to streamline operations and cash flows for most business firms.

In addition, two foreign studies show that such systems pose SME challenges in their implementation. Such studies highlight the need for well-designed, focused systems that balance functionality and ease of use rather than introducing unnecessary complexity that hinders the business.

The proposed system addresses these insights by focusing on the essential features that fit Module Pro's specific requirements, such as dashboards, payment reminders, and streamlined report generation. Excluding complex functionality like tax and payroll management, the proposed system ensures usability while tackling challenges such as manual tracking, payment delays, and inefficient data management. These studies

gathered by the proponents align with the company's objective and support its growth without overwhelming its operations.

METHODOLOGY

Kanban Agile Model

In developing an application, planning, research, and design are essential steps before moving to the development and publication phases. The Agile methodology provides a framework that outlines the process of software development. It follows a series of structured steps to ensure the creation of well-functioning software. A framework that falls under the Agile methodology is Kanban.

Kanban was developed in the late 1940s by a Japanese engineer named Taiichi Ohno. The Agile Kanban Framework focuses on visualizing the entire project on boards to increase project transparency and collaboration between team members. Kanban is one of the simplest frameworks used, as it allows project managers to efficiently manage and keep track of their projects. A distinguishing feature of the Kanban framework among different agile methodologies is its compatibility with the existing organizational setting (*see Appendix I, Figure 1, pg. 104*).

The proponents chose to utilize this model due to its adaptability, which aligns with the requirements of the system they are developing, a Project Management Information System. The Agile Model, with a chosen variant named Kanban, offers significant advantages, particularly in its iterative approach that allows developers to revisit and adjust previous stages as client requirements evolve. The model minimizes time wasted and reduces potential complications, as changes can be easily accommodated throughout the development process and tasks are divided into smaller tasks to create a sub-module or module of the system. Unlike the Scrum framework of the Agile Model, Kanban does not have any strict phases, allowing more fluid and continuous workflow; however, in a Kanban board, tasks are placed in different sections, providing information on what the task is currently at. The board provides the following common sections (*see Appendix I, Figure 1, pg. 104*):

To-Do Section

In this section, tasks such as client requirements and/or system features are placed. This section is where all the work to be done is organized before it moves to development. As requirements change, removal or editing of features or client requirements are done in this section. Tasks were also refined, adding in-depth details on what the task should be, in terms of functionality or design.

Doing/Development Section

Once the tasks are well-defined, they will now be moved into the “Doing” or “In Progress” section. This section is where the actual development takes place. Depending on the task, multiple tasks will be put in the section, depending on their difficulty, and some tasks will only be worked on at a given time or will be tagged as a Work In Progress (WIP) task, as the other tasks are then completed, to ensure efficient use of resources and time management.

Code Review/Testing Section

If the tasks were completed, they will now be moved into the “Testing” section and also align with the Code Review section. In this section, the tasks were tested and feedback occurred, having verification of whether the task met the requirements or its desired function. If that task does not meet the desired function or requirements, the task may then be put back into the Implementation phase, with a tag, to either fix the bug or change the task’s functionality to meet its requirements.

Deployment/Done Section

On the other hand, if the task has met its requirements, it will now be put into the “Deploy” or “Done” section. This phase is where all the finished tasks are placed and will be pushed to the system. Kanban allows all tasks to be reopened, changed, or improved if there are any sudden requirement changes or sudden bugs encountered when adding a new feature or requirement, maintaining the functionality of the application.

System design specification

- **Hardware/Software**

- **Hardware**

- **CPU**

- A minimum processor specification of either an Intel i3-6100 Dual-Core 3.7GHz or an AMD Ryzen 3 1200 Quad-Core 3.1GHz is recommended for developing the Project Management System. These processors, equipped with multiple cores, are capable of handling essential development tasks such as coding, testing, and database management. This specification ensures sufficient performance for multitasking operations throughout the development process.

- **Storage**

- A 500GB Solid-State Drive (SSD) is recommended for storage to support the installation of the Integrated Development Environment (IDE), database, operating system, and additional tools needed for system development. The SSDs' improved data processing, loading times, and code compilation speeds provide a smoother and more efficient development experience.

- **Memory (RAM)**

- A minimum of 8GB DDR4 2133MHz RAM is required, with 16GB DDR4 2400MHz recommended for optimal performance. This RAM capacity allows a smoother experience when running the IDE, multiple development tools, and a database concurrently, providing flexibility as project requirements expand.

- **Software**

- **Integrated Development Environment (IDE)**

- Visual Studio has been selected as the IDE for developing the Project Management Information System, using the C# programming language, Windows Forms, and .NET Framework. This setup supports efficient software development and offers robust tools for debugging, version control, and UI design.

- **Backend (Database)**

- SQL Server will serve as the database management system for the Project Management System, facilitating data storage, retrieval, and manipulation. It will act as the system's backend, handling critical operations such as data creation, editing, and deletion, essential for project management functionality.

- **Operating System**

- Windows 10 is chosen as the minimum operating system requirement. Its enhanced performance, stability, and compatibility with modern development tools and libraries make it an optimal choice for the development environment, ensuring efficient resource management and reducing potential compatibility issues.

RESULTS AND DISCUSSIONS

The developers efficiently provided access to the system, and each of the modules performed the anticipated tasks. In testing, the developers confirmed that the system operated accurately and performed those functions reliably, according to the client's operational processes. The proposed development was intended to enhance the major functions of the business, particularly regards scoping projects, tracking payments, and organizing project data.

The system improves the handling of scopes of work by enabling the admin to manage scope areas, items, and breakdowns through the Scope Settings module. This significantly reduces manual errors and ensures that scopes are standardized across projects. Staff can now easily select and apply existing scopes during project creation, speeding up the proposal process and avoiding redundancy.

The Projects module gives staff and admins a one-page view of all project records with status (pending, active, finished) depending on where the project is in the process. With this categorization, they can streamline project tracking and communication about the project's progress and focus on ongoing and upcoming priorities. The view for each project will also have the key details like the total material cost, delivery and installation fee, whether it is partially paid or fully paid, and other financial and essential details. The total amount paid for a project will be automatically adjusted by the system when the payment request is approved, so the staff neither has to manually track the total amount paid, reducing the chance of miscalculation.

The Requests module allows for communications between staff and the admin in terms of cancellation, payment requests, and completion approvals. The Payment requests add functionality for the staff to request payments against the project's remaining balance to ensure the project is not overpaid. Once the completion requests once approved, confirm that the project has been completed and provide a final proof of completion as documentation. The Cancellation Requests by the staff, when approved, confirm the cancellation of the project and provide the staff, as part of the request, the reason for cancellation, and a valid reason should be provided first, which must be confirmed as

valid by the admin. Furthermore, the admin can deny or reject the requests if the proof, reason, or payment is invalid and must require another evaluation.

The Staff Dashboard summarizes key information that helps staff understand overall project status and project selection preference, such as the most transacted scopes of work. Admins also have an Analytics module that shows some general metrics that provide a picture of performance, such as active, completed, and pending projects.

User access controls have also been placed so that only authorized users can conduct sensitive operations, i.e., modify scopes, change the status of projects, or approve items. This is a necessity in terms of securing sensitive operations and maintaining operational integrity (with respect to managing financial records and project documents).

The essential functions of the current system - project creation, payment tracking, managing scopes, and approval workflows - are successful and working towards the main objectives in the proposal, even though certain features are still in development (Notifications, Project Archiving, User Management, Payment Reminders).

Alpha testing was done by the developers internally to check that all the major modules were working as intended. The bugs and inconsistencies discovered during this testing were fixed to ensure the system was reliable and accurate. During beta testing, the client confirmed the features of the system in a simulated working environment but provided further feedback on usability and alignment with related processes. The results confirmed that the system performed to the expectations of the client in terms of working multiple project records, financial input management, and storage for completion documentation. However, the feedback provided (e.g., suggestions from the client and, in particular, the missing features) is accepted as part of the ongoing decision-making and will be used to further refine the system. This prepares the system for deployment and obtains optimal performance.

CONCLUSION

The developed Project Management Information System (PMIS) was designed to optimize the internal operations of the company, Module Pro Furniture and Fixture. It focuses on streamlining project creation, scope management, and payment tracking between admin and staff. The system replaces manual tracking processes—previously done through spreadsheets and logbooks—with an integrated interface that enables efficient recording, retrieval, and monitoring of projects and their financial progress. With a defined user access level, the system ensures proper validation and coordination of activities between the staff and admin, such as payment and completion approvals. Although the system is currently developed as a standalone application, it effectively supports the company's existing operational structure and workflow.

The system was evaluated through both alpha and beta testing, confirming that the core modules—project creation, payment tracking, completion approvals, and scope settings—function correctly. Admins can create and manage scopes, oversee projects, and approve requests, while staff can initiate new projects and request billing and completion approvals. The visual analytics provided to the admin allows for quick assessment of project statuses and completed works. Despite a few pending features (e.g., notification system, user management module), the system has proven to be functional and capable of handling the essential day-to-day operations of the business.

Based on the findings, the client acknowledged that the PMIS is aligned with the company's operational needs. The system enhances overall efficiency by digitizing processes, supporting structured project tracking, and improving data accuracy. This contributes to better organization, faster decision-making, and more consistent project execution within the company's scope of work.

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APPENDICES

APPENDIX A. GANTT CHART

MONTH	AUGUST				SEPTEMBER				OCTOBER				NOVEMBER				DECEMBER				JANUARY				FEBRUARY				MARCH				APRIL				MAY					
ACTIVITY																																										
Concepts and Requirements																																										
Identifying the problem																																										
Requirements gathering																																										
Review and analysis																																										
Design																																										
Architectural design																																										
Interface design																																										
Database design																																										
Program design																																										
Development																																										
System development																																										
Testing Phase																																										
System testing																																										
Beta testing																																										
Deployment																																										
Installation																																										
Maintenance																																										
System maintenance																																										

APPENDIX B. ACTUAL THESIS EXPENSES

THESIS EXPENSES

[illegible]

Prepared by:

Newton V. Ilagan	Raxelle V. Jumawan
Loise Mari V. Navarro	Ashanti R. Oriane

Noted by:

<Name of Parent/Guardian of Researcher>	<Name of Parent/Guardian of Researcher>
<Name of Parent/Guardian of Researcher>	<Name of Parent/Guardian of Researcher>

Approved by:

Christina T. Aseoche	Christina T. Aseoche
-----------------------------	-----------------------------

Budgetary Estimate

Quantity	Specifics	Approximate Cost
35 pages	Documents	₱105
3 pcs.	Folder	₱30
20 pages	Documents (objectives and scope and limitations part)	₱20

Human Resources

Christina Aseoche,
Professor,
STI College Fairview

Rimbert Romero,
CEO,
Module Pro Furniture & Fixture

APPENDIX C. USER's MANUAL

User's Manual

Insert here the PDF copy of your User's Manual. You may use one or more pages for your user's manual. Ensure that all necessary information is included

APPENDIX D. ADMIN'S MANUAL

Admin's Manual

Insert here the PDF copy of your User's Manual. You may use one or more pages for your user's manual. Ensure that all necessary information is included

APPENDIX E. CURRICULUM VITAE OF RESEARCHERS

Curriculum Vitae of
NEWTON V. ILAGAN
5-A Adrian St. Fairmont Subdivision North Fairview Quezon City
jrviarilagan@gmail.com
+639624880314

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	Sept. 2021 - July 2025	STI College Fairview
TechVoc	June 2018 - March 2021	STI College Fairview
High School	June 2014 - March 2018	Lagro High School
Elementary	June 2008 - March 2014	Lagro Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
n/a	n/a	n/a

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
n/a	n/a	n/a

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
Python Programming	5/10	2021
Video and Photo Editing	7/10	2019

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
n/a	n/a

Listed in reverse chronological order (most recent first).

Curriculum Vitae of
RAXELLE V. JUMAWAN
97 Santan St., Pingkian 3 Zone 3, Brgy. Pasong Tamo, Quezon City
jumawan@gmail.com
+639773232048

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	Sept. 2021 - July 2025	STI College Fairview
TechVoc	June 2019 - March 2021	STI College Fairview
High School	June 2015 - March 2019	Mary and Child Academy
Elementary	June 2009 - March 2015	Saint Francis Divine College (2009 - 2012) Mary and Child Academy (2012 - 2015)

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
n/a	n/a	n/a

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
April 2023 - July 2023	HIVE Society	Programming Committee

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
Video Editing	8/10	2017
Photo Editing	8/10	2018
C# Programming	6/10	2019
Game Development	6/10	2017
3D Modeling	7/10	2020
Music Production	5/10	2018

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
n/a	n/a

Listed in reverse chronological order (most recent first).

Curriculum Vitae of
LOISE MARI V. NAVARRO
B7 L11 Diamond Street, Dona Helen Subd. Camarin, Caloocan City
Loisenavarro58@gmail.com
+639295683000

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	Sept. 2021 - July 2025	STI College Fairview
TechVoc	June 2018 - March 2021	NU Fairview
High School	June 2014 - March 2018	St. Joseph College of Novaliches Inc.
Elementary	June 2008 - March 2014	Camarin D Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
n/a	n/a	n/a

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
n/a	n/a	n/a

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
Video Editing	6/10	2022
Microsoft Office	7/10	2010

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
n/a	n/a

Listed in reverse chronological order (most recent first).

Curriculum Vitae of
ASHANTI R. ORIANE
28 Kagawad Road Talanay Area A, Batasan Hills, Quezon City
orianeashanti21@gmail.com
+639163483856

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	Sept. 2021 - July 2025	STI College Fairview
TechVoc	June 2018 - March 2021	AICS Commonwealth
High School	June 2014 - March 2018	Batasan Hills National High School
Elementary	June 2008 - March 2014	San Diego Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
2020-2021	Content Moderator	Collective Solution

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
n/a	n/a	n/a

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
Python Programming	5/10	2021
Video and Photo Editing	7/10	2019

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
n/a	n/a

Listed in reverse chronological order (most recent first).

APPENDIX F: CALENDAR OF ACTIVITIES

Calendar of Activities

This section contains the detailed sequence of activities that the proponents will undergo in completing the thesis project while following the chosen methodology.

Week 1-3 (August 12-30, 2024)

To-Do:

- Orientation on the course curriculum and requirements.
- Formation of the proponents and brainstorming potential thesis topics.
- Identification of possible clients for the thesis.
- Assignment of roles and responsibilities to the proponents.

Doing:

- Creating the draft thesis proposal.

Done/Deploy (In the same week):

- Orientation on the course curriculum and requirements.
- Proponents were formed and brainstormed about the potential thesis topics.
- Identified the potential client: Module Pro Furniture and Fixture.
- Roles and responsibilities were assigned to the proponents.

Week 4 (September 1–6, 2024)

To-Do:

- Conduct an initial client interview to gather requirements.

Review:

- Submitted the draft of the thesis proposal to the thesis adviser for feedback.

Done/Deploy (in the same week):

- Conducted an initial interview with the client, Mr. Rimbart Romero.

Week 5-8 (September 9 – October 1, 2024)

Doing:

- Revision of the document based on the thesis adviser's feedback from Week 4.
 - Revision involved addressing concerns and clarifying details about the chosen client and project scope.

Done/Deploy (in the same week):

- The proposal was approved.
- Proponents were granted to seek an advisor, which they did, and later obtained a thesis adviser acceptance form, initiating the advisory phase.

Week 9 (October 6-12, 2024)

Doing:

- Consulting the thesis adviser to refine the project's goals and align with client requirements, and to receive advice about system suitability and documentation strategy.

Done (in the same week):

- Feedback and recommendations are incorporated into the project plan.

Week 10 (October 14-18, 2024)

To-Do:

- Conduct a formal client interview to gain a deeper understanding of their business processes.

Doing:

- Drafting Chapter 1:
 - Introduction
 - Background of the Problem
 - Overview of the Current State of Technology
 - Objectives of the Study

Done (in the same week):

- Conducted the formal client interview and received important information about the business process, which will help in the development of the system.
- Finished drafting Chapter 1 of the document.

Week 11 (October 21-27, 2024)

To-Do:

- Conduct another client interview for more structured information.

Doing:

- Creating the Methodology of the document.
- Outlining the Hardware and Software Requirements.

Review:

- Submitted the finished Chapter 1 to the adviser via Microsoft Teams for feedback.

Done (in the same week):

- Received more information about the company based on the conducted interview.
- Finalized the Methodology section.
- Outlined the Hard-/Soft-ware requirements.

Week 12-13 (November 2-8, 2024)

To-Do:

- Conduct another interview with the client for additional insights.

Doing:

- Revise the document based on the thesis adviser's feedback from Week 11, particularly:
 - Background of the Problem
 - Objectives of the Study
 - Methodology

- Creating the Review of Related Literature/Systems

Done (in the same week):

- Document revised
- Completed the RRL
- Insights obtained from the interview

Week 14 (November 11-15, 2024)

Doing:

- Add the necessary sections of the document:
 - Scope and Limitations of the Study
 - Appendixes
 - References
 - Transcript of Interview
 - Methodology Model
 - Human Resources
 - Curriculum Vitae
 - Budget Estimate

Done (in the same week):

- Added the necessary sections of the document.

Week 15-16 (November 18-29, 2024)

To-do:

- Fix Citations and Pages
- After revision, submit the document to the adviser for feedback and review.
- Perform another interview with the client
- Revise the Project Creation module
- Elaborate the Authorization and Security Access Module

Doing:

- Revising the Commonly Used Solutions in Overview of the Current State of the Technology
- Revising the Limitations of the Study
- Revising Backup Module
- Add a new module (requests by panelists)
- Add a new related system in Review of Related Literature/System (requests by panelists)
- Revise Calendar of Activities
- Revise the Scope of the Study
- Fixing citations and pages
- Alphabetize the References

Done (in the same week):

- Revised the Commonly Used Solutions in the Overview of the Current State of the Technology
 - Specified the solutions that Mr. Romero uses, and which are not being used in the business.
- Revised Limitations of the Study
 - Specified in Payment Processing that although the system does not accept transaction processing. The system will still accept cash only, this is for monitoring the transactions, which will be used as an indicator to show the status of a project financially, whether it regards budget, material purchases, and other relevant financial information.
 - Removed Forecasting limitation - will be put in the Recommendation part of the document.
 - Removed Tax Management and Payroll - it is a separate system; irrelevant to the study or proposed system.
- Revised the Backup Module
 - It is now the Backup and Restore Module
- Alphabetize the References

- Added SAP to Foreign-Related Systems in Review Related Literature/System
- Revised the Calendar of Activities:
 - Made sure that it follows the methodology as its guide.
- Revised the Scope of the Study
 - Added a Project Module
 - Specified the Scope by adding user-access levels and splitting who can access the modules.

Week 17-18 (December 2-10, 2024)

To-do:

- Pass the Soft and Hard copies of the document to the coordinator.

Doing:

- Performing another interview with the client
- Revising the Project Creation module
- Elaborating on the Authorization and Security Access module
- Fixing the citations and pages

Done (in the same week):

- (December 5, 2024) Passed the (non-final) soft copy to the adviser via Microsoft Teams for feedback and approval.
- (December 6, 2024) Performed the interview and added the answers to the Transcript of the Interview
- Revised the Project Creation module
 - Added the Client Address
 - Added the Project Type - whether the project is modular or renovation.
- Elaborated on the Authorization and Security Access Module
- (December 9, 2024) Fixed the citations and pages.
- (December 10, 2024) Submitted the final soft and hard copies of the document to the coordinator.

Week 19-20 (January 13-25, 2025)

To-do (This week):

- Create the User Info table
- UI Design
 - For Admin (Back-end)
 - For Staff (Front-end)
- Create windows/forms
 - Admin Window
 - Staff Window
 - Login Window
- Create, implement, and connect the Database to the system

Doing:

- Creating the user info table (Jan. 13)
- Creating windows (Jan. 14)
 - Admin
 - Staff
 - Login
- Minor UI designing (Jan. 14)
- Implementing the Login module (Jan. 17)
- Creating the database and table (Jan. 17)

Done:

- Created the system's database (Jan. 13)
- Created the User info table - tblUserInfo (Jan. 13)
- Created the windows of the following modules (Jan. 14):
 - Admin
 - Staff
 - Login
- Minor UI design for better readability (Jan. 21)
 - Made all the windows/modules' borders rounded for a modern look.

- Programmed the Login module (Jan. 21)
 - Insertion of Username
 - Insertion of Password
 - A login button that will direct the user to their respective modules: Admin or Staff.
- Successfully connected the database and the system (Jan. 24)

Week 21-22 (January 27 - February 8, 2025)

To-do (This week):

- Design the UI for Admin and Staff modules
- Add Functionalities to Admin and Staff Modules
- Start the Project Creation Module
- Create Tables

Doing:

- UI Designing and Adding Functionalities in Admin and Staff Module
- UI Designing in the Project Creation module

Done:

- Project Details Table (Feb. 2, 2025)
 - Created the following columns:
 - Project ID
 - Project Name
 - Project Description
 - Client Name
 - Client Phone Number
 - Client Telephone Number
 - Client Address
 - Client Email
 - Project Estimated Budget
 - Project Remaining Budget

- Project Type
 - Expected Start Date
 - Expected End Date
 - Project Status
 - Project Date Created
- Staff UI Design / Functionality (Feb. 1-4, 2025)
 - UI
 - Close
 - Maximize
 - Minimize
 - Active Projects
 - Updates
 - Payment Reminders
 - Personal Reminders*
 - Client's Info
 - Project Details
 - Create Project Button
- Created Flyout Navigation Menu:
 - Projects Tab
 - Clients Tab
- Functionality
 - Clicking "Create Project" will pop up a new windows form
 - Close Button
 - Close the system.
 - Maximize
 - Enlarge the system or restore it to its original size.
 - Minimize
 - Hides the system in the taskbar.
- Admin UI Design / Functionalities (Feb. 7-8, 2025)
 - Added the title of the form
 - Added a profile picture

- Added custom Windows control buttons:
 - Close
 - Maximize
 - Minimize
- Added Menu Buttons
 - Projects Button
 - Analytical Button
 - Settings Button
 - Logout Button
- Functionalities
 - Logout Button
 - Ask the user if they want to log out; they will be redirected back to the login form if so.
 - Close Button
 - Close the system.
 - Maximize
 - Enlarge the system or restore it to its original size.
 - Minimize
 - Hides the system in the taskbar.
 - The navigation menu switches the panel depending on what the user interacts with:
 - The analytics button directs the user to the analytics module
 - The Projects button directs the user to the Projects module
 - The manage users button directs the user to the manage users module
 - The settings button directs the user to the settings module

Week 23 (February 10 - 15, 2025)

To do:

- Continue the Project Creation Module (Functionality and Design)
- Re-design the Staff module

Doing:

- Project Creation Module Functionality (started Feb. 10, 2025)
- UI Re-design for Staff Page (started Feb. 13, 2025)
- Project Creation Module UI Re-design (started Feb. 13, 2025)

Done:

- Submitted a progress report video to thesis adviser (Feb. 11, 2025)
- UI Re-design for Staff Form (Feb. 14, 2025)
 - Re-designed window controls/buttons
 - Close Button
 - Maximize Button
 - Minimize Button
 - Re-designed and re-positioned the New Project button
 - Re-designed the menu and profile
- UI Re-designed for Project Creation module (Feb. 14, 2025)
 - Re-designed text fields and labels
 - Title (Module's header)
 - Project Name
 - Client Name
 - Client Contact Number
 - Client Address
 - Client Email
 - Project Starting Budget
 - Project Expected Start Date
 - Project Expected End Date
- Staff Form Functionalities (Feb. 15, 2025)
 - Adjusted and fixed minor errors in the sidebar menu.
- Project Creation Module Functionalities (Feb. 15, 2025)
 - Made sure that only numbers with special characters: '-' and '+', are only acceptable in the Contact Number input field.
 - The cancel button closes the window.

- Made sure that only numbers and decimals are accepted in the Project Starting Budget input field.

Week 24 (February 17 – 22, 2025)

To do:

- Continue the Project Creation module's functionality/design
- Re-design the Staff module

Doing:

- Project Creation Module Functionality (started Feb. 10, 2025)
- UI Re-design for Staff Page (started Feb. 13, 2025)
- Project Creation Module UI Re-design (started Feb. 13, 2025)

Done:

- UI Re-designed for Project Creation module (Feb. 16, 2025)
 - Re-designed the date field
- Project Creation Module Functionalities (Feb. 16, 2025)
 - Created a function that prevents the user from selecting an end date that is before the start date.

Week 25 (February 24 – 28, 2025)

To do:

- Re-design the Staff module
- Make the 'Active' projects in the Staff Dashboard update in real-time

Doing:

- UI Re-design for Staff Page (started Feb. 13)
- Making the active projects module in the Staff Dashboard update in real-time (Feb. 27)

Done:

- Fixed an error on the Expected End Date (Feb. 24)
- Made sure that the email address inputted is valid (Feb. 24)
- Project Creation functionality is made (Feb. 24)
- UI Design for Project Creation (Feb. 24)
- Shows an error when duplicated data, specifically the same – Project Name, Client Name, and Start Date, has been found in the database (Feb. 27)
- Made sure that all entries, in the Project Creation module, are valid, specifically: (Feb. 27)
 - Contact Number (min. 11 digits, no special characters, accepted characters [+ , -])
 - Email (must follow the format of an email [ex. username@mailserver.domain])
- Made sure that if a project has been created successfully, the Project Creation module will close. (Feb. 27)

Week 26 (March 2 – 8, 2025)

To-do:

- Continue re-designing the Staff module
- Continue the real-time data update in the ‘Active’ projects module
- Create a Data Grid View for showing tables

Doing:

- UI Re-design for Staff Page (started Feb. 13)
- Making the active projects module in the Staff Dashboard update in real-time (Feb. 27)
- UI Designing on the Data Grid View (Mar. 2)

Done:

- Some UI changes in Staff Dashboard (Mar. 2)
- Data Grid View added on Admin Dashboard: (Mar. 3)
 - Creating Projects on Staff updates the table on the Admin Dashboard – Projects tab
 - Making sure the necessary data is fully shown:

- Client Name
- Client Contact No.
- Initial and Remaining Budget
- Expected Start and End Date
- Project Status
- Project Date Created
- Made UI changes on the data grid view

Week 27 (March 9–15, 2025)

To-do:

- Continue re-designing the Staff module
- Re-design the Admin module
- Schedule an interview

Doing:

- Interviewing Mr. Romero (Mar. 9)
- Re-designing the Staff Dashboard (Mar. 10)
- Re-designing the Admin Dashboard (Mar. 11)

Feedback:

- Received Feedback on the first system check by the thesis adviser (Mar. 5)
- Received Feedback on an interview with the client (Mar. 9)

Done:

- UI Re-designing the Login module for a more modern look (Mar. 7)
 - Added an image, docked on the left side.
 - Changed some UI colors.
- The 'Enter' key submits the info in the Login module. (Mar. 7)
- Re-designing the Admin Dashboard (Mar. 11)
 - Removed the Profile
 - Removed the Payment Reminders Tab
 - Resized the window controls (min, max, and close)
 - Resized the Navigation buttons (Analytics, Projects, Settings, and Manage Users)

- Changed the Menu background (Mar. 12)

Week 28 (March 17–22, 2025)

To-do:

- Continue adding functionalities and design in the Admin module
- Schedule a meeting with the thesis adviser
- Create new tables

Doing:

- Making a dropdown "Admin" button containing: (Mar. 17)
 - Settings
 - Logout
- Asked adviser about a new update regarding the project management of the client (Mar. 18)
- Creation of new tables based on client's information about projects (Mar. 18)
 - tblProjectScope
 - tblMaterials
 - tblAccessories
 - tblUnitOfMeasures
- UI Re-designing in Admin module (Mar. 21)
- Fixing project creation module's errors and validation checkers (Mar. 21)

Done:

- Made the "Admin" button collapsible/a dropdown. Currently consists of the following: (Mar. 18)
 - Settings
 - Logout
- Made the following table: (Mar. 18)
 - tblProjectScope
 - scopeId (PK)
 - projectId (FK)
 - scopeName
 - scopeSpecs
 - scopeQty
 - scopeUom

- unitAmount
 - totalAmount
 - estCost
 - grossProfit
 - grossProfitPerc
- UI Changes on the Staff Dashboard (Mar. 19)
- UI Changes on the Login Module (Mar. 19)
 - Applied a new color scheme/palette
 - Applied the company's logo
- UI Changes on the Admin Module (Mar. 19)
 - Changed the font family and size of the dashboard.
 - Added the company's logo on the dashboard
 - Re-designed the tables' UI
- Fixed some validation checkers in the Project Creation module (Mar. 21):
 - The module can now accept null or empty inputs on the email text box.
 - The email validation error tooltip will now only pop up when the user begins typing or has entered a character.

Week 29 (March 24 - 29, 2025)

To-do:

- Continue re-designing the Admin and Staff module
- Fix some Staff and Admin functionalities

Doing:

- UI-designing in Admin Dashboard (Mar. 21)
- UI re-designing in Staff Dashboard (Mar. 25)
- Fixing/Changing Staff Module functionalities (Mar. 25)
- Fixing/Changing Admin Module Functionalities (Mar. 26)

Done:

- Re-designed the UI for the Staff Dashboard (Mar. 25)
 - Changed the UI so it matched the Admin Dashboard.
- Functionalities on Staff Dashboard
 - Changed that upon logging in as a Staff member, the Dashboard module will be automatically focused. (Mar. 25)

- Minor changes to window control for both Admin and Staff. (Mar. 26)
- Dragging the window while maximizing works as traditional windows were intended to be used.

Week 30 (March 30 - April 6, 2025)

To do:

- Fix functionalities in the Staff module
- Re-design the data grid view of the table tblProjectDetail
- Re-design the Login module
- Create/Modify tables
- Re-make the Project Creation module
- Continue re-designing the Staff and Admin module

Doing:

- Changing some functionalities in the Staff (Mar. 30)
- UI re-design of the tblProjectDetail's data grid view (Mar. 30)
- UI re-designing in the Login module (Mar. 31)
- Creating/Modifying database tables (Mar. 31)
- Re-making the Project Creation module (Apr. 1)
- UI re-designing in Staff Dashboard (Apr. 4)
- UI re-designing in Admin Dashboard (Apr. 4)

Done:

- Changed some functions in the Staff
 - Changed the Contact No. format from, "09", to "(+63)" (Mar. 30)
- Made changes in the UI design of the tblProjectDetail's data grid view
 - Made the column names' (header) font size bigger for readability (Mar. 30)
 - Changed the color palette to match the module color scheme (Apr. 4)
- UI designing in the Login module
 - Changed the BG color of the 'Login' button for consistency (Mar. 31)
 - Another color palette change (Apr. 4)
- Created/Modified database tables
 - Created tblProjectScope (Mar. 31)
 - ScopeId (PK) (ID seed = 20000000)

- ProjectId (FK from tblProjectDetails)
 - ScopeName
 - ScopeSpecs
 - ScopeQty
 - ScopeUom (Unit of Measure)
 - UnitAmount
 - TotalAmount
 - EstCost
 - GrossProfit
 - GrossProfitPerc
- Created tblProjectItems (Mar. 31)
 - ItemId (PK) (ID seed = 30000000)
 - ScopeId (FK from tblProjectScope)
 - ItemNo
 - ItemDesc
 - ItemQty
 - ItemCost
 - ItemTotalCost
- Modified tblProjectDetails (Mar. 31)
 - Changed the ProjectId seed from '2ID (Starting Id number) to '10000000'
 - Since the tblUserInfo already has the beginning ID of 1, more users create conflicts when it reaches more than 20.
- Created tblItems (Apr. 1)
 - ItemId (PK) (ID seed = 33000000)
 - ItemName
 - ItemCost
- Modified tblProjectItems (Apr. 1)
 - renamed 'ItemId' to 'ProjectItemId'
 - added ProjectID (FK from tblProjectDetails)
 - added ItemId (FK from tblItems)
- Project Creation Module Re-work
 - Made the module into a custom control rather than a pop-up form. (Apr. 1):
 - Page 1 (Project Details Module) UI Design (Apr. 1)
 - Project Name
 - Project Description
 - Client Name
 - Client Address
 - Contact Info (Contact No. and Email)

- Expected Start and End Date
- Initial Budget or Project Budget
- 'Next' Button
- 'Clear All' Button (Apr. 5)
- Page 1 Functionalities
 - Clicking the 'Clear All' button clears all input fields in the module. (Apr. 5)
 - Contact Numbers will always start at '(+63)' and have a limit of 15 characters, including the starting characters, '(+63)'. (Apr. 5)
- Page 2 (Project Scope of Works) UI Design (Apr. 6)
 - Added a dropdown list (Combo Box) containing the area of scope of a project.
- Milestone
 - Data can now be (temporarily) saved in a separate class. (Apr. 6)
- Fixes
 - Fixed the 'Next' button's position when the window is maximized. (Apr. 3)
 - Corrected the Expected End Date date time picker. (Wrong Spelling) (Apr. 5)
 - Made sure that the minimum date of the "Expected End Date" date time picker will not be able to be selected earlier than the "Expected Start Date". (Apr. 5)
 - The project budget text box should only accept decimal numbers. (Apr. 5)
 - The user can either leave the email text box empty, as it is optional. Alternatively, if the user wants to input an email, it must be valid. (Apr. 5)
 - Made sure that a project's start and end dates can't be the same, nor can the project start and end on the same day. (Apr. 5)
 - Before proceeding (clicking the 'Next' button), all necessary input fields must be valid and not null or empty (Apr. 6):
 - Project Name, Description, Client Name, Client Address, Contact No., and Budget must not be empty.
 - Contact No. must be valid (eg. (+63)9193056054)
- UI Design Changes in the Staff Module
 - Changed the color palette to a more 'appealing' tone (Apr. 4)

- UI Design Changes in the Admin Module
 - Changed the color palette to a more 'appealing' tone (Apr. 4)
 - Changed the default system icon to the company's logo (Apr. 4)

Week 31 (April 7 - 13, 2025)

To-do:

- Change functionalities in the Staff module
- Continue re-designing the Admin module
- Continue adding functionalities in the Admin module
- Fix some bugs

Doing:

- Changing some functionalities in the Staff (Mar. 30)
- UI re-designing in Admin Dashboard (Apr. 4)
- Adding functionalities in the Scope Settings module (Apr. 7)
- Bug fixing (Apr. 9)

Done:

- UI Design Changes in the Admin Module
 - Added a 'Scope Settings' button
 - This is where the admin can edit scope-related information. (i.e. Scope Areas, Scope of Work, and Items) (Apr. 7)
 - The following controls were created to complete the module:
 - Scope Area - the individual control for scope area (i.e. Kitchen) (Apr. 7)
 - The control (individual scope) contains the following controls: (Apr. 8)
 - Area Name Label
 - Edit Button
 - Delete Button
 - Insert Scope Area - the control for inputting the scope area. (Apr. 7)
 - The control contains the following controls:
 - Input field (inserting data/scope area)
 - Insert Button

- Cancel Button
 - Scope Area Panel - this is where all the individual scope area is shown, which are created by the Admin (Apr. 8)
 - Insert Items - the control for inputting the items in a scope. (Apr. 12)
 - The control contains the following controls:
 - Input fields
 - Item Name
 - Item Cost
 - Insert Button
 - Cancel Button
 - Scope Item - the individual control for the scope item and its respective price (i.e., Cabinet Hinges) (Apr. 13)
 - This control contains the following controls:
 - Item Name Label
 - Item Cost Label
 - Edit Button
 - Delete Button
 - Scope Item Panel - where the individual scope item is shown (Apr. 13)
- Added Functionalities in the Scope Settings module
 - 'Add Area of Scope' button: (Apr. 7)
 - Clicking the button shows a window containing an input field and two buttons - Insert and Cancel.
 - 'Cancel' asks the user for confirmation and then closes the window if confirmed.
 - 'Insert' takes the inputted data and inserts it into the table, tblProjectAreas. Duplicated or existing data will prompt the user to insert non-existing data.
 - Scope Area panel shows the individual scope inserted by the Admin. (Apr. 9)
 - A specific scope area contains the following (now functioning) button:
 - Delete button - deletes the selected scope area. (Apr. 9)
 - Edit button - now functional and can rename the selected scope area. (Apr. 11)
 - 'Add Scope Item' button: (Apr. 13)
 - Clicking the button shows a window containing two input fields and two buttons:
 - 'Cancel' asks the user for confirmation and then closes the window if confirmed.

- 'Insert' takes the inputted data - item name and cost. Then insert it into the table, tblScopeItems. Duplicated or existing data will prompt the user to insert non-existing data.
 - Scope Item panel shows the individual scope items inserted by the Admin (Apr. 13)
 - A specific scope item contains the following (now functioning) buttons:
 - Delete Button (Apr. 13)
- Created/Modified Database's table(s)
 - Created tblProjectAreas (Apr. 7)
 - AreaID (PK) (ID seed = 40000000)
 - AreaName
 - Renamed 'tblProjectAreas' to 'tblScopeAreas' (Apr. 13)
 - Renamed 'tblItems' to 'tblScopeItems' (Apr. 13)
- Functionalities in the Staff
 - Project Creation Module
 - The dropdown can now obtain the data from the table, tblProjectAreas. (Apr. 9)
 - This is where the Staff can choose what specific Area of Scope the project is going to be made.
- Fixes:
 - Fixed the tooltip position of the Email, Budget, and Contact Number textboxes; re-positioned them on the bottom-left side of their textbox boundaries (Apr. 9)
 - This will help the user to easily see and read what the tooltip is showing.
 - Fixed the incorrect message that is being shown in the tooltip in the Budget's textbox (Apr. 9)
 - Fixed the "CTRL+Backspace" feature of windows (Apr. 9)
 - Single-lined textboxes now feature the deletion per word/character when the user does the keyboard shortcut.
 - Fixed the width of the Scope Area. (Apr. 12)
 - Re-worked the Scope Area from Scope Settings in Admin (Apr. 12)
 - Changed the initial Table Layout Panel control to just a plain Panel.
 - Fixed the Edit button's control from Scope Area in Admin (Apr. 12)
 - When editing a scope area, instead of 'Insert', the button now says 'Rename'.
 - Fixes when adding a scope area:

- When the input starts with a whitespace, for example, “Kitchen”, the system will prompt a warning message if this is intended. They can rather choose yes or no if they want to continue and insert the data.
- Fixed duplicating workarounds:
 - Fixed checking duplicated with whitespaces (Apr. 12)
 - When inserting a new scope area, inserting ‘Bedroom’ when data like ‘Bedroom’ exists, the system will consider it as a duplicate canceling the data insertion operation.
 - Fixed checking duplicated with whitespaces when renaming (Apr. 13)
 - When renaming a scope area, for example, fixing a scope area from ‘LivingRoom’ to ‘Living Room’ will be allowed, however, it’s not allowed if it is the other way around. Accidental additional spaces, however, are allowed for fixing the wording.

Week 32 (April 14 - 20, 2025)

To-do:

- Continue to add and fix UI controls
- Continue to add functionalities
- Continue to add/fix the necessary tables in the database
- Fix some bugs and issues in the system

Doing:

- UI re-designing in Admin Dashboard (Apr. 4)
- Adding functionalities in the Scope Settings module (Apr. 7)
- Bug fixing (Apr. 9)
- UI designing in Staff Dashboard (Apr. 14)
- Changing the tables in the database (Apr. 18)

Done:

- Added/Changed UI controls:

- Changed the position of the buttons for adding scope items and areas: (Apr. 14)
 - For easier accessibility and more space to show the data.
- Added the following controls for the Scope of Works: (Apr. 14)
 - Panel Title (“Scopes of Work”)
 - Scopes of Work Control
 - The individual scopes
 - “Add Scope of Work” button
- Added the scope of work creation module UI which contains the following controls (Apr. 14):
 - Labels (Control Title, Scope Name Label, Scope Description Label, Items Title Label, Areas Title Label)
 - Input Fields
 - Scope Name
 - Scope Description
 - Scope Cost (Apr. 18)
 - Checklist Box
 - Areas
 - Items
 - Cancel button
 - Insert button
- Added an information label in the SoW creation module that states, “Scope’s total price may vary in the project creation module, where item quantities can be adjusted.” (Apr. 15)
- Added the scope’s total cost in the SoW creation module (Apr. 15)
- Changed the items’ checklist box into a table layout panel that consists of the following controls: (Apr. 17)
 - Checklist box
 - Item’s cost
 - Numeric Up/Down (Quantity changer)
 - Subtotal (Item’s cost X Quantity changer)
- Removed the information label (Apr. 17)
 - As the scope’s item prices are now being adjusted by the Admin instead of the staff
- Re-designed UI in Staff Dashboard (Apr. 19)
 - Made three (3) categories that consist of the following UI controls:
 - Active Projects

- Pending/”For Approval” Projects
 - Finished Projects
 - Each category’s projects consist of its basic details and a button to show the full details of the selected project.
- Functionalities Added/Changed:
 - A specific scope item contains the following (now functioning) buttons:
 - Edit Button (Apr. 14)
 - Pressing the ‘Enter’ and ‘ESC’ keys accepts and cancels the scope area/item creation/revision operation (Apr. 14)
 - Added functionality in the Login module where if the user presses the “ESC” key, the system asks the user for confirmation before closing the system (Apr. 14)
 - Clicking the “Add Scope of Work” buttons shows the dialog for creating a scope of work (Apr. 15)
 - Clicking the “Cancel” button in the Scope of Work (SoW) creation module will prompt the user for confirmation before closing the dialog box. (Apr. 15)
 - Scope Areas and Scope Items are now shown and updated in the SoW creation module. (Apr. 15)
 - The total (initial) cost of a scope now updates based on the checked/selected scope item (Apr. 15)
 - Feature of selecting items and modifying their quantities where the subtotal updates and the grand total of the items selected (Apr. 17)
 - Clicking the “View Full Details” button will show a window containing the project’s full details (Apr. 19)
- Database Changes:
 - Renamed all tables, column names, and the database to the standard naming convention (Apr. 18)
 - For easier understanding of the tables’ column names
 - Changed all tables’ ID seeds back to SSMS’ Identity (1, 1) (Apr. 18)
 - This is unnecessary since SSMS handles ID well
- Fixes:
 - Fixed an issue where ‘Cancel’ still closes the scope item/area creation module after pressing ‘No’. (Apr. 14)
 - Fixed the top panels (Scope Areas and Scope Items) so that it is equal to the bottom panel (Scope of Work) (Apr. 14)
 - Rephrased some confirmation messages for short yet concise understanding for the end-user (Apr. 18)

- In the Scope Area creation module, any input fields that start with a whitespace and/or end with a whitespace will be trimmed for easier data checking for duplication. (Apr. 18)
- Added an error message if the user wants to create a scope of work without a scope area or scope item (Apr. 20)
- The creation of a scope of work will not proceed until all fields and selections are valid (Apr. 20)
 - The scope's name, specification, and cost must be valid to proceed.
 - There should be at least one selected area and item

Week 33 (April 21 - 27, 2025)

To-do:

- Continue to add functionalities in the Scope Setting module
- Continue to add UI controls in the Scope Setting module
- Continue to fix bugs and errors
- Conduct a meeting with the thesis adviser

Doing:

- Adding functionalities in the Scope Settings module (Apr. 7)
- Adding UI controls in the Scope Settings module (Apr. 4)
- Adding tables in the database (Apr. 21)

Done:

- Tables Created/Modified (Apr. 21)
 - Table 'scopes'
 - Id (PK)
 - Name
 - Specs
 - Cost
 - Item_grand_total
 - Table 'scope_selected_areas'
 - Id (PK)
 - Scope_id (FK from table 'scopes')
 - Area_id (FK from table 'scope_areas')
 - Table 'scope_selected_items'

- Id (PK)
 - Scope_id (FK from table 'scopes')
 - Item_id (FK from table 'scope_items')
 - Quantity
 - Subtotal
- Added UI Controls (Apr. 21-22)
 - Added a "Scope" UI control
 - This holds the created scope and contains the following information:
 - Scope Name
 - Scope Cost
 - Scope Specs
 - Scope selected Areas
 - Scope Selected Items
 - Changed the icon of the delete button of scope, scope area, and scope item from a trash icon to an archive icon (Apr. 23)
- Added/Changed Functionalities (Apr. 21-22)
 - Scopes created by the admin are loaded in a panel in the Scope Settings module
 - The scope creation module is now functional (minus editing and deleting)
 - Deleting of scope area and item has now changed to Archiving (Apr. 23)
 - Archiving:
 - Adding an archived area/item will ask the user to restore it instead of creating a new (duplicated) one (Apr. 26)
 - Renaming an area/item to a previously archived name will prompt a restore and allow continued editing after restoration (Apr. 26)
- Conducted a meeting with the thesis adviser for system progress checking and recommendations (Apr. 23)
- Fixes
 - Minor UI re-positioning (Apr. 21-22)
 - Fixed the scope UI control's width (Apr. 25)
 - Fixed capitalization when editing an area (Apr. 26)
 - Fixed item price consistency (Apr. 27)
 - Item prices in scopes now remain fixed even if the original item's price is updated. The scope's total only changes when edited. Users will be notified of price updates, with options to either cancel, manually adjust prices, or create a new scope.

Week 34 (April 28 - May 3, 2025)

To-do:

- Continue to add functionalities in the Scope Setting module
- Continue to add UI controls in the Scope Setting module
- Continue to fix bugs and errors
- Add contents in Results and Discussion

Doing:

- Adding functionalities in the Scope Settings module (Apr. 7)
- Adding UI controls in the Scope Settings module (Apr. 4)
- Adding tables in the database (Apr. 21)
- Adding contents in Results and Discussion (May 2)

Done:

- Modified the column 'item_grand_total' in the table 'scopes'; added the following columns: (Apr. 28)
 - Delivery_installation_fee
 - Grand_total
 - Total_material_cost (fka item_grand_total)
- Added new UI labels in the Scope Creation module (Apr. 28)
 - Delivery/Installation Fee
 - Grand Total
 - Total Material Cost (fka Total Cost)
- The delivery and installation fee updates with the total material cost (formula: $0.15(\text{Customizable}) \times \text{Total Material Cost}$), as well as the total (delivery/installation fee + total material cost) (Apr. 28)
- Added a label in the panels of area, item, and scope that shows if there is no data in the panels (Apr. 30)
- Delivery/Installation fee and total now show accurate values in the scope's details (May 1)
- Archiving of the scope feature added (May 1)
- Fix renaming issues in the scope area and items to avoid duplication and improper error handling (May 1)
- Archived items, areas, and scope will not be shown in the scope settings unless restored (Apr. 26 - May 1)
- Functioning Scope edit module (May 2-3). Scope settings finished

Week 35 (May 5-10, 2025)

To-do:

- Rework the project creation module
- Add UI and functionality in the Settings module
- Continue to fix bugs and errors

Doing:

- Reworking the project creation module in the Staff module (May 6)
- Adding UI controls and functionalities in the Settings module (May 6)
- Minor UI fixing in the Scope creation module (May 6)
- Adding more content in Results and Discussion (May 2)

Done:

- Changed the font size of the delivery/installation fee in the Scope Creation module for readability (May 6)
- Admin can now change the delivery/installation fee percent in the Settings module (May 6)
- UI design in the Settings module (May 6)
 - Delivery/Installation fee percent textbox
 - Save button
 - Clicking will take the textbox's input and save it as a new delivery/installation fee percent.
 - Backup and Restore sub-module (May 6)
- Project Creation module changes/added:
 - Project Creation Module page 1 - selection of the project area - created (May 7)
 - Project Creation Module Page 2 - selection of the project's scope(s) (May 10)

Week 36 (May 12-18, 2025)

To-do:

- Continue to rework the project creation module
- Continue to fix bugs and errors

- Continue to add content to the Results and Discussion
- Add a customizable downpayment percent feature
- Add/Modify tables in the database
- Create the Project Approval Module
- Create the Engineers sub-module
- Conduct a meeting with the thesis adviser for progress checking

Doing:

- Reworking the project creation module in the Staff module (May 6)
- Adding contents in Results and Discussion (May 2)
- Adding/Modifying tables in database (May 12)
- Adding a customizable downpayment percent feature in the Settings of the Admin (May 12)
- Conducting a meeting with the thesis adviser for progress checking (May 14)
- Creating the functionality/UI design for the Engineer sub-module (May 17)
- Creating the Project Approval module (May 17)

Feedback:

- Conducted a meeting with the thesis adviser for progress checking of the system as well as the document (May 14)

Done:

- Finished putting contents in the Results and Discussion (May 12)
- Added/Modified tables in database (May 12)
 - Created table 'project_selected_scopes'
 - Id (PK)
 - Project_id (FK from table 'projects')
 - Name
 - Specs
 - Cost
 - Quantity
 - Subtotal
 - Modified the table 'projects'
 - Id (PK)
 - Name
 - Description
 - Client_name

- Client_phone_no
 - Client_address
 - Client_email
 - Exp_start_date
 - Exp_end_date
 - Total_scopes_cost
 - Total_items_cost
 - Overall_total
 - Downpayment
 - Progress_billing
 - Area (The area selected for this project)
 - Payment_status
 - Project_status
 - Date_created
 - Payment_status_date
 - Project_status_date
- Added a customizable down payment percent in the Settings module of the Admin (May 12)
- Fixed the input field of contact info in Page 3 of the Project Creation module (May 13)
- Project Creation Module done (May 13)
- Created a module that will pop up before submitting for approval (May 13)
- Removed the items' total in all modules in the Project Creation Module (May 16)
- Fixed/Changed the calculation of the amount to be paid (May 16)
- Projects are now shown in the Projects Module (May 16)
- Minor UI changes in the Project Creation Module and Projects Module for readability (May 16)
- Scope Control UI and Functionality Changes
 - Made the items sorted alphabetically when editing/creating a scope (May 16)
 - Changed the items' total color (May 16)
- Changed the way delivery/installation fee percent and downpayment/progress billing fee percent are saved (May 16)
 - Created a new table for saving the fee percentages (table 'fee_settings')
 - Id (PK)
 - Delins_fee_percent
 - Downpayment_percent
 - Quotation_validity (May 18)
 - Max_leadtime (May 19)
- UI Changes in the Admin Module

- Changed the 'Projects' button into a dropdown button and added categories (May 16)
 - Pending (For Approval)
 - Active
 - Finished
 - All
 - Cancelled
 - Archived
- Add a 'Logs' button in the 'Admin' dropdown button (May 16)
- Added an 'engineer' table (May 17)
 - Id (PK)
 - Name
 - Is_busy
 - Is_archived
- Engineer Functionality (May 17)
 - Adding/Archiving Engineer function implemented in the Projects Settings (FKA Scope Settings)
- Created a module for accepting 'For Approval' projects in the 'Pending' projects tab in Admin Module (May 17)
- Added a project card in the 'Pending' tab to show the basic details of a project. On click, a breakdown form and a button that accepts the project and changes the status from 'For Approval' to 'Active' (May 17)
- Added a new control that is visually similar to the project breakdown module before submitting for approval. The difference is it contains admin-only information (i.e., Items included, Total Items Cost, Gross Profit, etc.) (May 17)
- Changes in the Project Creation Module Pg. 3 (May 18)
 - Staff can no longer set the project starting date and ending date of a project, due to the company's terms and conditions
 - Put the selected Area on the 3rd page for reminding purposes
- Added Quotation Validity on 'For Approval' projects (May 18)
 - The remaining days of validity for a project is customizable in the Settings module of the Admin

Week 37 (May 19-25, 2025)

To-do:

- Continue to add UI designs and functionalities across the system
- Continue to fix bugs and errors found in the system
- Conduct another meeting with the thesis adviser

- Continue to modify/add tables in the database

Doing:

- UI designs and functionalities of the system (May 19)
- Fixing bugs and errors found in the system (May 19)

Done:

- Minor Fixes
 - Changed the “Scopes of Work” into “Scope of Work” in the Project Settings (May 19)
 - Fixed the width of the area in the Scope sub-module of the Scope of Work in Project Settings (May 19)
 - Fixed the inputs in the Settings module (May 25)
 - Fixed the delivery/installation fee calculation logic (May 25)
- Reworked the Approval Form (May 19)
 - Changed the Exp. Start Date into a DateTimePicker control
 - Expected End Date is now calculated based on the max. Leadtime set by the admin plus the Expected Start Date.
 - Added the Engineer dropdown list.
 - Added the Project’s Payment Status
- Added UI/Functionalities
 - Added the customizable functionality of the maximum lead time in the Admin’s Settings module (May 19)
 - Project Approval in Admin’s Pending Projects module is now functional (May 19)
 - Active projects are now shown in the Admin’s Active Projects Module (May 19)
 - Reworked the Staff Dashboard module (May 20)
 - Created Project Cards and Project Forms (May 20)
 - Created the Staff Request Module (May 23)
 - Created the Admin Request Approval Module (May 23)
 - Complete project workflow implemented (May 24)

- The number of requests is now shown and updated in real time in the Admin module (May 25)
- Database Changes
 - Added table 'requests' (May 20)
- Passed an overall progress check and demo to the thesis adviser for checking (May 24)

Week 38 (May 26 - May 31, 2025)

To-do:

- Continue to add UI designs and functionalities across the system
- Continue to fix bugs and errors found in the system
- Continue to modify/add tables in the database
- Add contents to the document

Doing:

- UI designs and functionalities of the system (May 19)
- Fixing bugs and errors found in the system (May 19)
- Database changing (May 20)
- Adding contents to the document (May 21)
- Conducting an interview with the client (May 26)

Feedback:

- Conducted an online interview with the client and has received answers regarding the additional information about the business processes (May 29)
- Conducted the System Demo and obtained feedback from the panelists (May 30)

Done:

- Minor Fixes
 - Fixed the instantiating logic of Pending Projects in Staff Dashboard (May 26)
 - Changed the logic on how completion proofs are saved (May 27)
 - Made the date formats consistent - MMM dd, yyyy (May 27)

- UI Fixes (May 27)
- Added UI/Functionalities
 - Added Suggested Scope Cost based on the desired gross profit (found in the Admin's Settings Module) when inserting/editing a scope (May 26)
 - Active projects are now being instantiated in the Staff Dashboard, sorted by the start date of the project (May 26)
 - Added a request counter in the Admin module (May 27)
 - Top selected scopes are now being instantiated in the Staff Dashboard (May 27)
 - Request Rejection is now functional (May 27)
 - Fixed a bug where a request can still be made when a pending project is fully paid (May 27)
- Database Changes
 - Changed the data type of 'completion_proof' from varchar to varbinary (May 27)
- Added Contents to the Document:
 - Results and Discussion (May 29)
 - Conclusion (May 29)
 - Inserting the sample project data provided by the client (May 29)
 - Fixing the pages in the table of contents (May 29)

Week 39-40 (June 1 - 14, 2025)

To-do:

- Rework Staff Dashboard

Doing:

- Reworking the Staff Dashboard; both UI and Functionalities (June 8)

Done:

- Reworked the Staff Dashboard (June 14)

Week 41 (June 15 - 21, 2025)

To-do:

- Minor rework in the Projects Module
- Separate the information of the customer from the project.

Doing:

- Reworking the Staff Dashboard; both UI and Functionalities (June 14)
- Reworking the Projects Module; both UI and Functionalities (June 16)
- Changing some UI and Functionalities in the 3rd page of the Project Creation Module (June 17)
- Changing some UI and Functionalities in the Projects Module, Project Details Module, and other project breakdown modules (June 21)

Done:

- UI Design Changes, Fixes, and Additions
 - Reworked the Projects Module (June 16)
 - Minor UI changes in the Project Creation module - Page 3 (June 20)
 - Features auto-complete on the client information text box.
 - Added 'RESET' button on the client's name text box.
 - Changed the Project Name into a label, instead of a text box, to introduce the auto-generating project name (June 21)
- Functionalities Added, Fixed, and Removed
 - Double clicking a cell row will now open the project details in the reworked Staff Dashboard (June 15)
 - Search and filter are now functioning on all Project modules (June 16)
 - Project Names are now auto-generated following this naming format: PRJ-[Project ID shown in 4 Digits]-[Client's Initials]-[Project Area] (June 21)
 - Fixed how projects, dashboard (in Staff), project breakdown information, and requests are loaded, to comply with the client-project info separation (June 21)

- Created a new table, “customers”. This will split the information creating and updating from the customer to the project (June 17)

Week 42 (June 22 - 28, 2025)

To-do:

- Start working on the Analytics Module
- Start working on the Users Module
- Bring back the Attention/Engineer UI and its functionalities

Doing:

- UI/Functionality implementation in Analytics Module (June 22)
- Bringing back the attention/engineer UI and functionalities (June 24)
- UI/Functionality implementation in the Users Module (June 25)

Done:

- UI Design Changes, Fixes, and Additions
 - UI for Analytics Module (June 23)
 - Brought back the Attention Panel in the Project Settings (June 24)
 - Added the contact label (June 24)
 - Added an input field for contact in the Attention Creation Sub-module (June 24)
 - Project Approval Module now contains the engineer/attention dropdown (June 24)
 - UI for Users Module (June 25)
 - UI for User Password Changer Sub-module (June 27)
 - UI for User Edit Sub-module (June 28)
 - UI for User Creation Sub-module (June 28)
- Functionalities Added, Fixed, and Removed
 - Implemented all-time project status count, time period selection menu, actual gross profit line chart based on selected time period, and project

status distribution donut based on selected time period in the Analytics Module (June 23)

- Attention now has contact information saved (June 24)
- User (Admin in particular) must now select an available attention/engineer in the dropdown menu of the Project Approval Module (June 24)
- Searching functionality, filtering functionality, and Loading of users added in the Users Module (June 25)
- User archiving functionality (June 25)
- User Password Changer Sub-module functionality implemented (June 28)
- User Edit Sub-module functionality implemented (June 28)
- User Creation Sub-module functionalities implemented (June 29)
- Database Changes
 - Added a new column in the ‘engineers’ table named ‘contact_no’ (June 23)

Week 43 (June 29 - July 5, 2025)

To-do:

- Fix logic on the Analytics Module
- Add Archiving features
- Continue working on the Users Management Module
- Fix the “auto-cancel” function when validity date is reached
- Add printing functions

Doing:

- UI/Functionality implementation in the Users Module (June 25)
- Adding Archiving features (UI/Functionality) (June 30)
- Fix issues and bugs in the system (June 30)
- Adding the Print functions (July 2)

Done:

- UI Design Changes, Fixes, and Additions
 - Added UI for Archived Projects (June 30)

- Functionalities Added, Fixed, and Removed
 - User Creation Sub-module functionalities implemented (June 29)
 - Added Archiving/Restoring function on Admin Project Details (June 30)
 - Fixed the logic in Analytics so that it only shows the necessary values (June 30)
 - Fixed the issue where reached validity date are not being automatically cancelled (June 30)
 - Added printing feature on Admin's Project Details (July 2)

Week 44 (July 6 - July 12, 2025)

To-do:

- Add printing functions
- Overhaul Project Details
- Continue to bugfix

Doing:

- Fix issues and bugs in the system (June 30)
- Adding the Print functions (July 2)
- Overhauling Project Details - both Admin and Staff (July 6)

Done:

- UI Design Changes, Fixes, and Additions
 - Overhauled the Project Details for Admin (July 8)
 - (In print preview) Changed/Fixed the sizing logic of Proof of Completion (July 10)
 - (In print preview) Included the scope's cost (July 10)
 - Overhauled the Project Details for Staff (July 10)
 - Printing Module for Both Admin and Staff Project Details added (July 10)
 - Project Creation Module Pg. 3 Minor Changes (July 10):
 - Clicking the 'Reset' button will also clear all client-related input fields

- Hid a debug message
- Fixed the Quotation Validity Date format in Project Proposal Form/Module (July 10)

Week 45 (July 13 - July 19, 2025)

To-do:

- Continue to fix bugs and errors
- Add Receipt Generation sub-module for Admin
- Change UI and Logic of Payment Request Sub-module
- Make changes in the database
- Make a minor logic change on how projects are instantiated
- “Modernized” the printing preview feature

Doing:

- Fix issues and bugs in the system (June 30)
- Adding the Receipt Generation sub-module (July 13)
- Minor UI and Logic adjustments in Payment Request Sub-module (July 13)
- Making changes in the database (July 13)
- Making logic changes on Project instantiation (July 15)
- Changing the printing preview logic in both Admin and Staff Project Details module (July 16)

Done:

- UI Design Changes, Fixes, and Additions
 - Modified the Payment Request Sub-module to contain radio buttons on first-time payment projects and a manual input for the progress billing (July 13)
 - Added a new selection in the Request drop-down menu - “Request Project for Delivery” - non-functional. (July 13)
 - Added remarks text box on Request Payment Update sub-module (July 15)
 - Added remarks label on Request Payment Update Approval sub-module (July 15)

- Functionality Changes, Fixes, and Additions
 - Prepared the logic for the Receipt Generation sub-module (July 13)
 - Minor logic change on how payment requests are made (July 13)
 - Payment tracking implemented (but not yet shown) (July 15)
- Database Changes
 - Added the table 'project_payments' (July 13)

Week 46 (July 20 - July 26, 2025)

To-do:

- Continue to fix bugs and errors
- Add Receipt Generation sub-module for Admin
- “Modernized” the printing preview feature

Doing:

- Fix issues and bugs in the system (June 30)
- Adding the Receipt Generation sub-module (July 13)
- Changing the printing preview logic in both Admin and Staff Project Details module (July 16)

Done:

- Functionality Changes, Fixes, and Additions
 - Printing logic will now automatically open the default browser to preview the PDF file of the Project Details - both Admin and Staff (July 20)
 - Receipt generation added (July 21)

Week 47-48 (July 27 - August 9, 2025)

To-do:

- Continue to fix bugs and errors
- Payment Reminder Module

Doing:

- Payment Reminder Module (Aug 3)

Done:

- UI Changes:
 - Created a 'Create Reminder' button in the Admin Project Details (Aug 3)
 - Created a sub-module UI for the Payment Reminder Creation Module (Aug 3)
 - Created a module UI for the Payment Reminders Module (Aug 3)
 - Buttons in the Admin Project Details Module are now icon-based rather than text-based (Aug 3)
- Logic Changes:
 - Payment Reminder Creation Module partial completion (Aug 4)
 - Generated reminders can now be sent thru the user's email
 - Created reminders are now properly created in the database
 - Added a 'remarks' input field (Aug 7)
 - Payment Reminders Module partial completion (Aug 7)
 - Reminders are now being read and loaded in the UI
 - Searching and filtering of reminders added
- Database Changes:
 - Created/Updated 'reminders' table (Aug 3)
 - Added 'created_by' column (Aug 4)

Week 49-50 (August 10 - 23, 2025)

To-do:

- Continue to fix bugs and errors
- Continuation of Payment Reminder Module

Doing:

- Payment Reminders Module (*since* Aug 3)

Done:

- UI Changes:
 - Status in the Payment Reminders Module have their own respective color (Aug 11)
 - Admin Project Details Module (Aug 18)
 - Total Items Paid
 - Items Remaining Balance
 - Project Remaining Balance
- Logic Changes:
 - Payment Reminders Module (Aug 11)
 - Admins can now edit reminders
 - Automated status change of 'Overdue' payments
 - Fixed the bug when extending 'Overdue' payments remains 'Overdue'
 - A pop-up message will now appear for the amount of overdue payments
 - Admins can now re-send a payment reminder to their email
 - Admins can now mark a reminder as 'Paid' (Aug 18)
 - Paid reminders will now reflect its amount to the 'materials_amount_paid' column
- Database Changes:
 - Created 'reminders' table (Aug 3)
 - Added 'created_by' column (Aug 4)
 - Updated 'projects' table (Aug 18)
 - Added 'material_amount_paid' column

Week 51-52 (August 24 - September 6, 2025)

To-do:

- Continue to fix bugs and errors
- Continuation of Payment Reminder Module

- Portfolio Module

Doing:

- Payment Reminders Module (*since* Aug 3)
- Portfolio Module (Sep 1)

Done:

- Milestones:
 - Payment Reminders Module completed (Aug 24)
 - Portfolio Module completed (Sep 1)
- UI Changes:
 - Added the Portfolio Module UI (Sep 1)
 - A dedicated ‘UI-card’
 - Overhauled the UI for Scope Area Creation Module (Sep 4)
- Logic Changes:
 - Added a reversion of payment in Payment Reminders Module (Aug 24)
 - Portfolio Module (Sep 1)
 - Searching, Sorting, and Filtering
 - Clicking its custom card will open more info
 - Image is now required when adding an area in Scope Area Creation Module (Sep 4)
 - Fixed the engineers logic where their ‘is_busy’ column is not being updated (Sep 4)
- Database Changes:
 - Added ‘image’ column in ‘scope_areas’ (Sep 4)

Week 53 (September 7 - 13, 2025)

To-do:

- Continue to fix bugs and errors
- Overhaul Scope of Work Creation/Update Module
- Overhaul Scope Settings Module

Doing:

- Scope of Work Creation/Update Module (Sep 11)
- Scope Settings Module (Sep 7)

Done:

- UI Changes:
 - Change the UI control for: (Sep 7)
 - Scope Items
 - Scope of Work
 - Engineers/Attention
 - Created a Photo Card UI control (Sep 7)
- Logic Changes:
 - Areas and Items are now being loaded in the new UI control of SoW Creation/Update Module (Sep 11)
 - Refactored and Optimized the Photo Card UI Control (Sep 12)
 - Image uploading, proper image ordering, and proper removing are now properly operating
- Database Changes:
 - Created table 'scope_images' (Sep 7)

Week 54 (September 14 - 20, 2025)

To-do:

- Continue to fix bugs and errors
- Continue overhauling Scope of Work Creation/Update Module
- Continue overhauling Project Settings Module

Doing:

- Scope of Work Creation/Update Module (*since* Sep 11)
- Project Settings Module (*since* Sep 7)

Done:

- Milestones:
 - SoW Creation/Update Module completed (Sep 18)
- Logic Changes:
 - SoW Creation/Update Module
 - Duplicate/Archive checking implemented (Sep 15)
 - Includes restoring option if archived
- Fixes:
 - Payment Reminders Module (Sep 18)
 - Fixed the message when reverting a payment reminder
 - Fixed the issue that archived payment reminders still have options besides ‘restore’
 - Fixed the issue when restoring an archived payment reminder, the UI doesn’t load properly
 - Payment Reminder Creation Module (Sep 18)
 - Fixed that when clicking ‘no’ when reverting, it still creates a reminder
 - Fixed the issue where e-mail is not being sent
 - SoW Creation/Update Module (Sep 18)
 - Cost input field should now accept numbers only
 - Optimized the loading times when opening the Project Settings Module (Sep 18)

Week 55 (September 21 - 27, 2025)

To-do:

- Continue to fix bugs and errors
- Continue overhauling Project Settings Module
- Update/Fix Project Creation Module
- Custom SoW Request Module

Doing:

- Project Settings Module (*since* Sep 7)
- Project Creation Module (Sep 25)
- Custom SoW Request Module (Sep 26)

Done:

- UI Changes:
 - Buttons in the Page 1 (Area Selection) Project Creation Module are now resized properly (Sep 25)
 - Images are now being shown in the Area Selection Sub-Module
- Database Changes:
 - Created table 'scope_requests' (Sep 26)

Week 56 (September 28 - October 4, 2025)

To-do:

- Continue to fix bugs and errors
- Continue overhauling Project Settings Module
- Continue to Update/Fix Project Creation Module
- Continue Custom SoW Request Module
- Overhaul Requests Module

Doing:

- Project Settings Module (*since* Sep 7)
- Project Creation Module (*since* Sep 25)
- Custom SoW Request Module (*since* Sep 26)
- Requests Module (Sep 30)

Done:

- Logic Changes:
 - SoW Requests are now being loaded in the Requests Module (Oct 1)
 - SoW Approval Module (Oct 3)

- Navigation of multiple requested scopes are now functional
 - Data are consistent when navigating
 - UI Changes:
 - Overhauled the UI of Requests Module into a more user-friendly UI (Sep 30)
 - SoW Approval Module added (Oct 1)

Week 57 (October 5 - 11, 2025)

To-do:

- Continue to fix bugs and errors
- Continue overhauling Project Settings Module
- Continue to Update/Fix Project Creation Module
- Continue Custom SoW Request Module
- Continue to overhaul Requests Module
- For Evaluation and For Delivery Module

Doing:

- Project Settings Module (*since Sep 7*)
- Project Creation Module (*since Sep 25*)
- Custom SoW Request Module (*since Sep 26*)
- Requests Module (*since Sep 30*)
- For Evaluation Module (Oct 5)
- For Delivery Module (Oct 6)

Done:

- Milestones:
 - Workflow for Made-to-Order Projects are now completed (Oct 9)
- Logic Changes:
 - Clicking the image in a Photo Card will now enlarge it (Oct 5)
 - Rejection of a Requested Scope implemented (Oct 5)

- Project selected scope images are now being saved on a separate table for a consistent project history (Oct 7)
- Requests in Requests Module can now be opened by double-clicking (Oct 8)
- Project Timeline is now being updated each approved requests (Oct 9)
- UI Changes:
 - Added buttons and the sub-module for 'Delivery' and 'Evaluation' projects in Admin Module (Oct 6)
 - Added a new Project Details for Projects Under Evaluation (Oct 6)
 - Added a "Delivery/Installation Ready" photo proof in both Admin and Staff Project Details Module (Oct 6)
 - Added the "Del./Ins. Ready" photo in the Print Generation Module (Oct 6)
 - Added a new UI control to show a custom/requested scope of work (Oct 6)
 - Added the buttons of 'Delivery' and 'Evaluation' in the Staff Module (Oct 7)
 - Added a dropdown for Staff before logging out (Oct 7)
 - All Scope of Work UI Controls now have 2- to 3-button navigation
- Database Changes:
 - Created table 'project_timelines' (Oct 5)
 - Added column 'installation_proof' in table 'projects' (Oct 6)
 - Added table 'project_selected_scope_images' (Oct 7)

APPENDIX G: TRANSCRIPT OF INTERVIEW

1. Can you please tell (us) the nature and operations of your business, sir, Romero?

Gumagawa po kami ng mga Modular cabinets, furnitures, and fixtures, start-up company pa po to and mag-dadalawang taon na siyang operational, and ang ginagawa namin ino-*offer* namin ang services namin sa paggawa ng mga custom na gamit halimbawa, mga renovation sa bahay, mga kitchen or kung ano man gusto ng kliyenteng ibago sa kanilang bahay.

2. Ilan po yung mga empleyado niyo sa company?

Sa ngayon ako lang since startup pa nga lang and ayoko panaman maghawak ng mga tao pagdating sa business ko, pero pag sabihin na nating may *client* nako at may pinapagawa, dun lng ako maghi-hire ng mga taga gawa, tulad ng mga capentero, mga taga pintura, ako na din bahala sa sweldo nila, baon nadin at transpo- yung iba kase kelangan pa ng transpo- fee para makapunta.

3. Pano niyo po nahahandale mga paperworks since kayo lang po nagmamanage ng mga documents and payments?

Mano-mano, sa monitoring kahit papano, sinusulat ko nalang muna sa noteboook ko, minsan sa Excel ‘pag sinisipag, pati din sa payments ni-re-record ko nalang din sa Excel. At tsaka dahil iba-iba mga *client*, hinahati-hati ko sila per project per client kaya mahirap ‘pag ‘di ko kaagad naayos sa Excel.

4. Ano po yung mga main informaton na kelangan i-monitor para sa management ng business niyo, ano po yung mga key information para mag-run smoothly yung business niyo?

Yung budget na binigay ng *client*, payroll nadin ng mga empleyado, expenses and mga payments na kailangan bayarin; yung mga binibiling materials at mga baon ng mga tauhan.

5. When it comes sa management niyo po sa business, ano po yung most problematic sainyo or saang pong part kayo nahihirapan?

‘Pag dating sa recording minsan nakakalimutan ko i-record, halimbawa, may several projects dalawa or tatlong projects, na-o-off track na ‘ko, tulad ng mga nagastos, lalo na sa materials na ginagamit sa paggawa, yung payments din minsan ‘di ko kaagad nababayaran, kaya kelangan talaga maalala ko or ma-remind ko sarili ko na, kumbaga kelangan ko bayaran to ngayong araw o sa susunod. Kunwari, pagdating sa isang project,

may budget na binibigay si client, since yun may mga babayaran at bili ng materials, ‘pag ‘di ko kaagad na-record yun ‘di ko agad nacocompute, para kahit papano nakikita ko kung mag kano na nagastos ko sa project na ‘to, lalo na kasi yung matitira sa budget, yun na yung commision kung baga.

6. Ano po yung mga step-by-step process ng project kapag may client na?

A-ask ko muna kung ano papagawa, if mga modular lang or renovations. Ask ng budget, depending sa papagawa, bibigay ng timeline.

7. Follow-up question po sa step-by-step process with the client. Ibig sabihin po ba kapag may scope of work kayong, let’s say, di’ kayang magawa, the entire project will be cancelled?

Bago ko kasi i-encode, fina-finalize ko muna with the client. Kung isang scope of work is unable to produce, either renegotiate or hindi na maco-continue.

8. Ano pong mga data/details ang dapat naka-enter sa paggagawa ng projects?

Needed data and details (like) client name, contact information, project address, materials costs, at project budget.

9. Ano yung mga details ang kinukuha niyo sa client?

Client name, contact info, project concept/idea, ga’no kalaki yung work area.

10. Sa type ng paggawa, dalawa lang po ba?

The main product/service is Modular cabinets. Additional offer na lang po yung mga fit-out since pareho naman home improvement; same market.

11. With regards sa work area, what do you mean by that? Is it a name-like format, for example, *kitchen area* or *living room area*, or is it more specific measurements like *15m²* or *4x2m room*?

Customized modular cabinets. Kung ano sukat and design ng client, yun po ang maaring gawin.

12. When it comes to tasks na naka-assign, do you also have a list of people and specific tasks na naka-assign doon sa tao na yon? How does it work?

Yes, may specific person for cutting ng cabinets: critical sa’min yan. Others are helper na lang po.

13. For the materials, how do you track them? What information or details do you put?

Materials-wise, when the system is implemented, ano po yung mga expected niyong makita?

Sourcing of materials are just from local hardwares or specialty supplier. I am just expecting na ma-track ko ang cost of materials each project.

14. Based po dun sa Excel na binigay niyo. This is only for one specific project, correct?

Yes, materials buy for specific project.

15. How many days po ang pag-gather ng mga requirements niyo for one project? Does it take a whole day to complete the scope of works, as well as the materials needed?

2-3 (days) production to installation lead time.

16. Is it fixed? For example, once these materials and requirements are placed in the Excel sheet, no further additions needed?

Yes, fixed na po ito 'pag na-install.

17. Pa'no yung process ng pagkuha ng customer? Do they call you? Thru messenger? Sa FB page? Face-to-face?

They mainly reach out sa FB page or Viber pero eventually, mag-me-meet face-to-face for agreement and finalization.

18. Where do you usually get these clients po?

Since small business palang kami, most ng mga clients is within Quezon City and parts ng Metro Manila.

19. Ano po ang inyong return/refund policy? This is a follow up question about sa “cancellation of any portions of this contract subject to forfeiture of down-payment for the orders placed and produced”.

No return policy because most are customized. Shop drawings are being presented prior to production.

20. (Follow up from Q17) What about the sa mga paid progress billing, particularly yung portion?

I haven't experienced it yet but once the client decided not to pursue the contract. the assumption is everything delivered, or work rendered should be paid or should be collected.

21. Was there ever a scenario na when buying the materials for a project, iba yung cost niya sa Excel file, or any tool na ginagamit niyo for tracking items in the scope of work, compare sa actual (potentially updated) item cost? For example, in the Excel file na binigay niyo po, the Edge Band costs Php27.00. However, nung bibilhin niyo na po, it's actual cost na is, let's say, Php30. May ganyang scenario na po ba kayong na-experience? Or lagi po kayong updated sa item costs? Or this scenario po ba is impossible to happen?

Progress billing usually works on bigger projects or depending on negotiations. The percentage of progress billing is to be determined by both client and supplier.

22. Do some customers po ba do walk-ins?

We do not have an actual store, rather an office. We get clients thru referrals and online.



CEO,
Module Pro Furniture and Fixture

APPENDIX H: DOCUMENTED IMAGES



MODULE PRO
FURNITURE FIXTURE

Modular Cabinets *Customized Furniture *Joineries *Toilet Partitions *Interior Fit-out
*Office Partitions and Furniture *Window Blinds and Curtains

Office Address: 115 Villareal st., Bo. Gulod, Novaliches,
Quezon City
Contact nos.CP: 0906-4552611
Landline: 8707-0406

PROJECT NAME : _____ **No.** **25-028**

PROJECT LOCATION : Mandaluyong City

ATTENTION : Engr. Aaron Gatbonton

RE: Supply, delivery and installation of Customized Cabinets

DATE: March 5, 2025

We are pleased to submit our quotation for the above mentioned requirements;

Scope of works	Specifications	Qty.	UOM	Unit Amount	Total Amount	EST. COST	GP						
Cabinet A Hang Cabinet	Cabinets Carcass: Marine ply Melamine Laminated; Door Panels : Marine Ply in Laminated Finished; Mechanism: Bluware Soft Close Hinges	1	lot	22,000.00	22,000.00	15,191.50	6,808.50 31%						
Cabinet A Sink Cabinet	Cabinets Carcass: Marine ply Melamine Laminated; Door Panels : Marine Ply in Laminated Finished; Mechanism: Bluware Soft Close Hinges	1	lot	23,000.00	23,000.00	14,731.50	8,268.50 36%						
Cabinet B Hang Cabinet	Cabinets Carcass: Marine ply Melamine Laminated; Door Panels : Marine Ply in Laminated Finished; Mechanism: Bluware Soft Close Hinges	1	lot	25,000.00	25,000.00	16,939.50	8,060.50 32%						
Cabinet B Sink Cabinet	Cabinets Carcass: Marine ply Melamine Laminated; Door Panels : Marine Ply in Laminated Finished; Mechanism: Bluware Soft Close Hinges	1	lot	22,000.00	22,000.00	14,708.50	7,291.50 33%						
Cabinet C Hang Cabinet	Cabinets Carcass: Marine ply Melamine Laminated; Door Panels : Marine Ply in Laminated Finished; Mechanism: Bluware Soft Close Hinges	1	lot	32,000.00	32,000.00	22,080.00	9,920.00 31%						
Cabinet C Sink Cabinet	Cabinets Carcass: Marine ply Melamine Laminated; Door Panels : Marine Ply in Laminated Finished; Mechanism: Bluware Soft Close Hinges	1	lot	24,000.00	24,000.00	15,019.00	8,981.00 37%						
Note: Countertop OSM <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 40%;">Total</td> <td>148,000.00</td> </tr> <tr> <td>VAT</td> <td></td> </tr> <tr> <td>Grand Total</td> <td></td> </tr> </table>						Total	148,000.00	VAT		Grand Total		98,670.00	49,330.00 33%
Total	148,000.00												
VAT													
Grand Total													

Not included:
Back Splash, Sink, appliances, fixtures, plumbing / electrical / gas line connection / termination, demolition of existing kitchen, subdivision fees, bonds, parking fee and the likes, and other items not mentioned above.

TERMS AND CONDITIONS:

*All items delivered shall remained a property of MODULE PRO FURNITURE AND FIXTURE TRADING. until fully paid by client.
*Please make check payable to MODULE PRO FURNITURE AND FIXTURE TRADING
*Quotation validity: 15 days
*Mode of payment: 65% downpayment; 35% progress billing until completion.
*Leadtime: 30-45 days upon confirmation/issuance of PO and receiving of down-payment.
*Cancellation of any portion of this contract is subject to forfeiture of down-payment for the orders placed and produced.
*WARRANTY: 3 months for parts and services.
*Warranty is not applicable to damaged due to misuse; civil or carpentry works damage
NON-DISCLOSURE: Any information contained in the proposal shall be treated as confidential and shall not be disclosed to anyone without consent of MODULE PRO FURNITURE AND FIXTURE TRADING

Prepared by:

Rimbert Romero
General Manager

Authorized signature over printed name
This serves as Purchase Order

Image 1: A single project details (The right side is for the admin - Mr. Romero)

ITEM NO.	Cabinet A Hang	ITEM DESCRIPTION	QTY	COST	TOTAL COST
1		Marine Laminated	2	2,500.00	5,000.00
		18mm Cutting	2	1,200.00	2,400.00
		6mm MARINE ply 4x8" Laminated Single face	1	1,250.00	1,250.00
		6mm Cutting	1	400.00	400.00
		HPL	1	1,200.00	1,200.00
		HPL PRESSING	1	400.00	400.00
		hinges	6	125.00	750.00
		Edge band	30	27.00	810.00
		Adhesives	1	300.00	300.00
		Screws and tox	1	100.00	100.00
		adjustable plinth		100.00	-
		Grab Handle	1	600.00	600.00
		Drawe guide			-
					-
					-

TOTAL MATERIAL COST 13,210.00
delivery/installation 1,981.50
TOTAL COST 15,191.50

ITEM NO.	Cabinet A Sink	ITEM DESCRIPTION	QTY	COST	TOTAL COST
1		Marine Laminated	2	2,500.00	5,000.00
		PVC		3,750.00	-
		18mm Cutting	2	1,200.00	2,400.00
		6mm MARINE ply 4x8" Laminated Single face	1	1,250.00	1,250.00
		6mm Cutting	1	400.00	400.00
		HPL	1	1,200.00	1,200.00
		HPL PRESSING	1	400.00	400.00
		Hinges	2	125.00	250.00
		Aluminum		3,000.00	-
		Edge band	30	27.00	810.00
		Adhesives	1	300.00	300.00
		Screws and tox	1	100.00	100.00
		adjustable plinth	1	100.00	100.00
		Grab Handle	1	600.00	600.00
		Drawe guide		400.00	-

TOTAL MATERIAL COST 12,810.00
delivery/installation 1,921.50
TOTAL COST 14,731.50

ITEM NO.	Cabinet B Hang	ITEM DESCRIPTION	QTY	COST	TOTAL COST
1		Marine Laminated	2	2,500.00	5,000.00
		18mm Cutting	2	1,200.00	2,400.00
		6mm MARINE ply 4x8" Laminated Single face	1	1,250.00	1,250.00
		6mm Cutting	1	400.00	400.00
		HPL	1	1,200.00	1,200.00
		HPL PRESSING	1	400.00	400.00
		hinges	12	125.00	1,500.00
		Edge band	40	27.00	1,080.00
		Adhesives	1	300.00	300.00
		Screws and tox	1	100.00	100.00
		adjustable plinth		100.00	-
		Grab Handle	1	1,100.00	1,100.00
		Drawe guide			-
					-
					-

TOTAL MATERIAL COST 14,730.00
delivery/installation 2,209.50
TOTAL COST 16,939.50

Image 2: Material Details for a single project (more in-depth details from Image 1)

ITEM NO.	Cabinet B Sink	ITEM DESCRIPTION	QTY	COST	TOTAL COST
1		Marine Laminated	2	2,500.00	5,000.00
		PVC		3,750.00	-
		18mm Cutting	2	1,200.00	2,400.00
		6mm MARINE ply 4x8" Laminated Single face	1	1,250.00	1,250.00
		6mm Cutting	1	400.00	400.00
		HPL	1	1,200.00	1,200.00
		HPL PRESSING	1	400.00	400.00
		Hinges	4	125.00	500.00
		Aluminum		3,000.00	-
		Edge band	20	27.00	540.00
		Adhesives	1	300.00	300.00
		Screws and tox	1	100.00	100.00
		adjustable plinth	1	100.00	100.00
		Grab Handle	1	600.00	600.00
		Drawe guide		400.00	-

TOTAL MATERIAL COST 12,790.00
delivery/installation 1,918.50
TOTAL COST 14,708.50

ITEM NO.	Cabinet C Hang	ITEM DESCRIPTION	QTY	COST	TOTAL COST
1		Marine Laminated	3	2,500.00	7,500.00
		18mm Cutting	3	1,200.00	3,600.00
		6mm MARINE ply 4x8" Laminated Single face	1	1,250.00	1,250.00
		6mm Cutting	1	400.00	400.00
		HPL	1	1,200.00	1,200.00
		HPL PRESSING	1	400.00	400.00
		hinges	16	125.00	2,000.00
		Edge band	50	27.00	1,350.00
		Adhesives	1	300.00	300.00
		Screws and tox	1	100.00	100.00
		adjustable plinth		100.00	-
		Grab Handle	1	1,100.00	1,100.00
		Drawe guide			-
					-
					-

TOTAL MATERIAL COST 19,200.00
delivery/installation 2,880.00
TOTAL COST 22,080.00

ITEM NO.	Cabinet C Sink	ITEM DESCRIPTION	QTY	COST	TOTAL COST
1		Marine Laminated	2	2,500.00	5,000.00
		PVC		3,750.00	-
		18mm Cutting	2	1,200.00	2,400.00
		6mm MARINE ply 4x8" Laminated Single face	1	1,250.00	1,250.00
		6mm Cutting	1	400.00	400.00
		HPL	1	1,200.00	1,200.00
		HPL PRESSING	1	400.00	400.00
		Hinges	4	125.00	500.00
		Aluminum		3,000.00	-
		Edge band	30	27.00	810.00
		Adhesives	1	300.00	300.00
		Screws and tox	1	100.00	100.00
		adjustable plinth	1	100.00	100.00
		Grab Handle	1	600.00	600.00
		Drawe guide		400.00	-

TOTAL MATERIAL COST 13,060.00
delivery/installation 1,959.00
TOTAL COST 15,019.00

Image 3: 2nd page of material, Details for a single project (more in-depth details from Image 1)

APPENDIX I: KANBAN AGILE METHODOLOGY

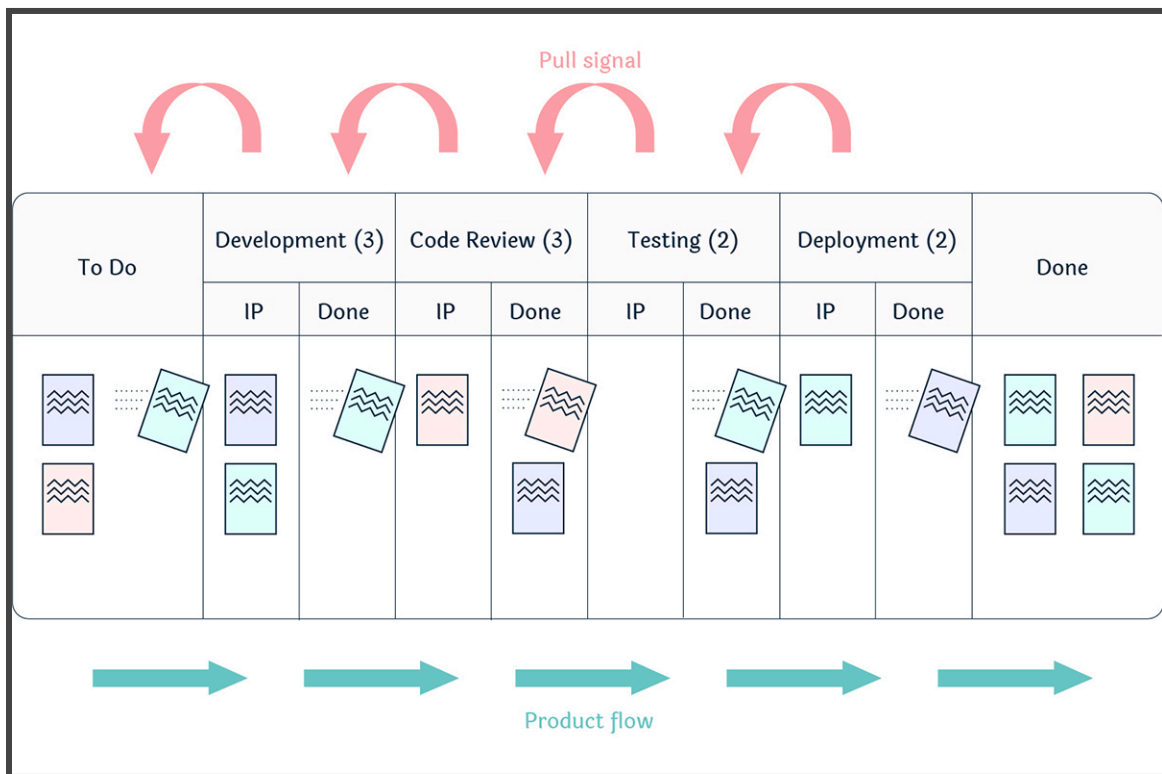


Figure 1: Kanban Agile Methodology Model

ADVISER'S ACCEPTANCE FORM

NAME OF PROPONENTS:

Newton V. Ilagan
Raxelle V. Jumawan
Loise Mari V. Navarro
Ashanti R. Oriane

APPROVED RESEARCH TITLE:

Project Manager Information System for Module Pro
Furniture and Fixture

AREA OF STUDY:

Project Management Information Systems (PMIS)
Business Process Management

CONFORME:

Christina T. Aseoche
Thesis Adviser

APPROVED BY:

Christina T. Aseoche **Date:** 17 October 2024
Thesis Coordinator

NOTED BY:

Leodigario R. Bongco
Program Head